



Imbalanced data based fault diagnosis of the chiller via integrating a new resampling technique with an improved ensemble extreme learning machine

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ABSTRACT

Fault diagnosis of the chiller is essential to guarantee chiller's safe operation and reduce building energy consumption. However, the existing fault diagnosis methods rarely consider the chiller's imbalanced data conditions, which always leads to low diagnosis accuracy of the minority class samples. To figure out the issue of imbalanced fault pattern data during the chiller fault diagnosis, a hybrid resampling based improved extreme learning machine (HRIELM) is developed in our work. A new hybrid resampling technique (HRT) is first proposed to balance the majority and minority classes in the imbalanced fault pattern datasets. The HRT is then carried out repetitively to obtain multiple diverse rebalanced training datasets using different benchmark datasets. Subsequently, various primary extreme learning machine (ELM) models are established utilizing these rebalanced training datasets. Furthermore, an improved ELM model based on a novel selective ensemble learning strategy is presented to enhance the effectiveness of the chiller fault diagnosis. The basic ELM models with superior performance for diagnosing the neighbor subset constructed according to the fault snapshot sample are first picked out, and a weight factor is further defined to build the final selective ensemble ELM model. Finally, the fault pattern of the snapshot sample is identified according to the weighted voting mechanism. Detailed experimental results on the ASHRAE Research Project RP-1043 experimental datasets certify the effectiveness of the presented HRIELM scheme for the chiller fault diagnosis under the imbalanced data environments.

1. Introduction

As an important component of the heating, ventilation, and air-conditioning (HVAC) system, the chiller can greatly influence the overall running status of the HVAC [1]. If the chiller with faults is not maintained timely, the control strategy of the chiller will become invalid which would deteriorate the HVAC's operating effectiveness. When fault diagnosis techniques are adopted to repair the faults occurring in the chiller, the energy consumption of the HVAC system can be reduced by about 30% [2]. Therefore, the fault diagnosis of

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chiller is of great significance to reduce the building energy consumption and increase the building comfort. Recently, the research on fault detection and diagnosis of chiller systems has gradually become one of the hot issues. However, for most data-driven based methods, the models are usually trained on the balanced fault pattern datasets of the chiller, which results in poor fault diagnosis performance for the minority class samples [3].

The data imbalance phenomenon among different chiller's fault patterns mainly results from the troublesome issue of gathering insufficient fault pattern samples as well as some specific fault patterns possess much higher occurrence frequencies than that of other fault patterns [4,5]. The imbalanced datasets refer to at least one fault pattern subset (the minority class) that is usually outnumbered by the other fault pattern subsets (the majority class). Specifically, the detailed reasons which give rise to the problem of the chiller's data imbalance are analyzed in Section 3.2. Whereas, the traditional chiller fault diagnosis models are constructed on the balanced fault pattern datasets by maximizing the overall diagnosis accuracy. When the traditional fault diagnosis models are employed to recognize the imbalanced fault snapshot datasets of the chiller, the minority class samples with greater interest than the majority class samples may be misclassified in most cases. Therefore, an improved fault diagnosis method is urgently needed to enhance the fault diagnosis accuracy of the minority classes. As the two typical data sampling level methods, the synthetic minority over-sampling technique (SMOTE) based over-sampling [6,7] and the random under-sampling [8,9] possess complementary advantages and shortcomings to deal with the problem of imbalanced data. Therefore, it is essential to come up with a new data resampling technique by incorporating the SMOTE based over-sampling with the random under-sampling, for the purpose of figuring out the chiller's data imbalanced problem.

After the imbalanced datasets of the chiller are rebalanced, the data-driven based fault diagnosis techniques are usually employed to efficiently analyze the data. Currently, the extreme learning machine (ELM) based fault diagnosis algorithms have attracted more and more attention by virtue of their fast learning speed and good generalization performance [10,11]. Nonetheless, the fault identification accuracy and robustness of the single ELM model still need to be reinforced. Fortunately, the ensemble learning strategy has been applied to enhance the accuracy and robustness of the single ELM based fault diagnosis models by integrating a large number of basic ELM models [12,13]. However, the conventional ensemble learning based ELM fault diagnosis approaches have some shortcomings. On the one hand, when the number of integrated basic ELM models increases, it becomes more and more difficult to distinguish the performance differences among the numerous basic ELM diagnosis models. On the other hand, the more basic ELM diagnosis models are utilized, the greater the computation complexity of the final ensemble model will possess. In order to tackle the aforementioned defects of the traditional ensemble learning algorithms, a lot of studies about the selective ensemble learning approaches have been conducted and reported [14–16]. Inspired by the advantages of selective ensemble learning, we put forward an improved ELM fault diagnosis model to identify the chiller's faults after the historical fault pattern datasets are rebalanced, where a new selective ensemble learning scheme is infused into the standard ELM algorithm.

In our work, to figure out the above mentioned tough issues, a new hybrid resampling based improved ELM (HRIELM) diagnosis scheme is presented to efficiently recognize the faults of the chiller under the imbalanced fault pattern data environments. To be specific, a new hybrid resampling technique (HRT) is first constructed to rebalance the imbalanced fault pattern datasets of the chiller by incorporating the synthetic minority over-sampling technique (SMOTE) with the random under-sampling. On the basis of the rebalanced fault pattern datasets, an improved selective ensemble learning based ELM fault diagnosis strategy is further developed to recognize the patterns of fault snapshot datasets by effectively choosing the multiple basic ELM models with superior diagnosis accuracies and performance. Our main contributions are given out.

- (1) To rebalance the imbalanced historical fault pattern datasets of the chiller, a novel HRT approach is presented by integrating the SMOTE over-sampling method with the random under-sampling method. A benchmark dataset is first determined from the C kinds of imbalanced fault pattern datasets. The SMOTE over-sampling method is then employed to enlarge the minority datasets whose numbers of the samples are smaller than that of the benchmark dataset by generating the artificial synthetic samples, while the random under-sampling method is utilized to shrink the majority datasets whose numbers of the samples are larger than that of the benchmark dataset by randomly eliminating the samples. Therefore, the C kinds of the imbalanced fault pattern datasets are rebalanced to have the same number of samples as that of the benchmark dataset.
- (2) To accelerate the training process and reduce the computational burden during the chiller fault diagnosis, the standard ELM which owns fast learning speed and excellent generalization performance is employed as the basic fault diagnosis model. To be specific, every imbalanced dataset of the C kinds fault patterns is regarded as the benchmark dataset, hence the developed HRT method is repetitively performed to acquire a total of C different rebalanced training datasets, each of which also involves C classes balanced fault pattern datasets. Afterward, to lay a solid foundation for the subsequent selective ensemble learning based fault diagnosis scheme, the number of C various basic ELM diagnosis models are respectively constructed on these C different rebalanced training datasets.
- (3) To further strengthen the fault diagnosis effectiveness of the chiller and alleviate the computation complexity, a modified selective ensemble learning strategy is established by distinguishing the different diagnosis capabilities of the multiple basic ELM models trained on the diverse rebalanced training datasets. The nearest neighbors of the fault snapshot sample are first searched in the verification dataset to establish the neighbor subset. After the basic ELM models with the superior fault diagnosis performance for the neighbor subset are picked out, a weight factor is further defined to construct the final ensemble ELM based fault diagnosis model. At last, the pattern of the fault snapshot sample is identified by the developed selective ensemble ELM model according to the weighted voting mechanism.
- (4) Detailed experiments of the proposed hybrid resampling based improved ELM (HRIELM) based fault diagnosis scheme is carried out on the ASHRAE RP-1043 datasets, and abundant comparisons with the interrelated fault diagnosis methods are given out.

The remarkable capability of the developed HRT to handle the chiller's imbalanced data is first proved in our experiments. Afterward, the experiments and comparisons exhibit that the presented HRIELM scheme achieves outstanding fault diagnosis effectiveness and reveals significant performance improvements in comparison with the interrelated approaches.

The structure of this paper is formulated as: Section 2 gives out a detailed literature review of our work, while Section 3 introduces the operating principle of the chiller and analyzes the problem of the chiller's data imbalance in detail. The developed HRT algorithm is described elaborately in Section 4. The presented selective ensemble learning based ELM diagnosis model is comprehensively constructed in Section 5. Section 6 provides the suggested HRIELM based fault diagnosis scheme for the chiller under imbalanced data conditions. Detailed experimental results are carried out in Section 7. Finally, the conclusions are provided in Section 8.

2. Background

In recent years, various fault diagnosis techniques have been applied to the HVAC system to guarantee efficient building energy utilization and favorable occupant comfort [17,18]. The existing fault diagnosis techniques are primarily classified into model [19,20] and data-driven based approaches [21–23]. As a typical fault diagnosis technique, the model based approaches have been studied by many researchers to maintain the HVAC system running normally. For instance, based on the P&ID control strategy and the model architecture, Taal et al. [24] developed a novel diagnostic Bayesian network to recognize the faults of the HVAC system automatically. Fan et al. [25] extracted the practical physical operation information from the factory installed sensors to establish the modified support vector machine (SVM) based diagnosis models for recognizing the chiller's seven typical faults. By integrating an advanced Bayesian nonlinear state estimation technique with a back-smoothing method, Bonvini et al. [26] proposed an operational mechanism based model to detect the energy and hydraulic faults of the chiller. However, the model based fault diagnosis methods commonly have difficulty in acquiring quantitative physical knowledge of the operation status of the chiller, air handling unit (AHU), and other components of the HVAC system. Besides, the performance of the model based fault diagnosis methods is often affected by the strong nonlinearity and time-varying dynamics of the chiller. Thus, these model based approaches would result in the unsatisfied fault diagnosis performance for the complex, large-scale HVAC system.

A large number of data-driven based approaches have been suggested to diagnose the faults of the chiller recently. For example, with the help of adapting decision boundaries, Van et al. [27] combined the transfer component analysis and the SVM to transmit the knowledge among different chillers. By treating the fault diagnosis task as a multi-class classification problem, Li et al. [28] suggested an improved linear discriminant analysis method to realize the chiller fault diagnosis. Liu et al. [29] proposed a new association algorithm based fault diagnosis method to extract the operational rules of the chiller by association rule mining. In addition, we have conducted some studies to detect and recognize the faults of the chiller and AHU by combining the slow feature analysis based approaches with the deep learning based approaches [30–34]. Based on the above reviewed literature, it can be found that the data-driven based approaches have been extensively utilized to identify the chiller fault. However, during the actual running of the chiller, historical fault pattern datasets are often imbalanced, which simultaneously contain the majority class datasets and the minority class datasets. Under the chiller's imbalanced data environments, these data-driven based approaches have the common shortcoming that they tend to maximize the overall fault diagnosis accuracy while ignoring the chiller's minority class fault patterns [3].

To strengthen the diagnosis effectiveness under the imbalanced data conditions, the studies from the levels of data sampling [35], cost-sensitive learning [36], bagging, and boosting based ensemble [37] have been reported in recent years. At the data sampling level, over-sampling and under-sampling are the most popular approaches to tackle the problem of imbalanced data. The under-sampling and over-sampling approaches transform the imbalanced datasets into the balanced datasets by removing the majority class samples or augmenting the minority class samples. As a classical over-sampling method, the synthetic minority over-sampling technique (SMOTE) and its improved versions are becoming more and more popular. Particularly, by combining the SMOTE with random search algorithm and gradient boosting tree method, Chen et al. [6] presented an improved ensemble classifier to achieve efficient and accurate rock trace identification. To consider the distribution information of the training data and exploit its internal structure, Zhang et al. [7] discussed an instance weighted SMOTE method with enhanced robustness by indirectly utilizing the data distribution. Although the SMOTE based over-sampling approaches are widely utilized to handle the imbalanced data, the existing methods may suffer from the limitations of overlapping synthetic samples and the distribution marginalization, i.e., the boundary between majority class samples and minority class samples is blurred. Therefore, the improvements for the SMOTE based over-sampling methods need to be made to generate high quality samples to augment the minority classes.

Random under-sampling is another extensively adopted algorithm to settle the issue of data imbalance at the data sampling level. The random under-sampling algorithm rebalances the data samples in majority and minority classes by randomly decreasing the samples in majority classes according to the number of samples in a benchmark class. Lu et al. [8] suggested a new fault diagnosis method to distinguish the motor bearings' operation conditions by under-sampling the vibration signals. In order to improve the prediction of extracellular matrix proteins, Kabir et al. [9] utilized a modified asymmetric under-sampling approach to avoid the class imbalance problem, and then adopted the SVM to perform cross validation for each dataset. Nevertheless, because the datasets are rebalanced by eliminating some samples from the majority classes, the random under-sampling algorithm may ignore a large amount of useful and meaningful information of the majority classes.

Based on the rebalanced datasets of the chiller, the ELM based fault diagnosis algorithms have attracted more and more attention due to their fast learning speed and good generalization performance [10,11]. For instance, to improve the diagnosis performance of the vibration signal, Wang et al. [38] put forward an optimized extreme learning machine by integrating with an adaptive weight

particle swarm optimization model. Yan et al. [39] combined the modified Kalman filter with an improved ELM model to diagnose the AHU system fault by classifying the time series. Based on an enhanced kernel ELM with the random features representation, Liu et al. [12] presented a new network based method to identify the misuse of computer systems. Although various applications have displayed the decent diagnosis results of the ELM based approaches, the single ELM model's diagnosis accuracy and robustness are need to be enhanced.

By fusing multiple basic ELM models, the ensemble learning strategy has been discussed to improve the performance of the single ELM based fault diagnosis models [12–14]. Nevertheless, with the increment of the integrated basic ELM models, it becomes more difficult to distinguish the basic ELM diagnosis models' performance differences, and the final ensemble model will possess the greater computation complexity. In recent years, the selective ensemble learning approaches have been studied to handle the shortcomings of the traditional ensemble learning algorithms. For example, by building a novel soft selection strategy to weight several machine learning models, Lv et al. [14] established an efficient model combination process and evaluated the validation accuracy of the model ensemble. To deal with the data's dynamic characteristics changing along with time, Jin et al. [15] discussed a modified forecasting method using the selective ensemble to perform wind power forecasting. Li et al. [16] suggested a beetle antennae search algorithm based on an improved selective ensemble technique to diagnose the rolling bearing faults. Therefore, based on the advantages of selective ensemble learning, an improved ELM fault diagnosis model is urgent to be established for recognizing the chiller's faults.

3. The operating mechanism and imbalanced data of the chiller

To promote understanding of the presented imbalanced data based fault diagnosis scheme for the chiller, the operating mechanism of the chiller is first explained and the data imbalance phenomenon is then revealed.

3.1. Introduction of the chiller

As the major constituent part of the HVAC, the chiller is to adjust the temperature of the room through the circulation of the refrigerant. Fig. 1 gives out the refrigerant flow path in the chiller [4,32]. As displayed in Fig. 1, The chiller system is formed by four basic components, which are the compressor, condenser, evaporator, and throttle valve. Among them, the compressor is the core component of the chiller, and its function is to provide energy for the circulation of the chiller. The condenser and evaporator are two heat exchangers which responsible for the heat exchange between the chiller system and the environment. The throttle valve connects the condenser and the evaporator, which can reduce the pressure and the temperature of the liquid in the refrigeration pipeline. To sum up, the chiller continuously exchanges heat with the indoor and outdoor environments by the flow of refrigerant in the major components.

In the actual running of the chiller, the duration of normal operation is much longer than that of the operation with faults. Even the faults arise, some specific fault patterns own much higher occurrence frequencies than that of other fault patterns. Therefore, the sufficient fault pattern samples are always unavailable and the numbers of different fault pattern samples are usually imbalanced, i.e., at least one fault pattern dataset (the minority class) is usually outnumbered by the other fault pattern datasets (the majority class).

3.2. The chiller's data imbalance problem

At present, more and more researchers have employed the data-driven based approaches to diagnosis the faults of the chiller. However, in the practical scenarios, the problem of the chiller data imbalance severely affects the fault identification accuracies of the minority fault patterns, thus having a negative impact on the traditional fault diagnosis method [40,41]. The phenomenon of data imbalance is caused by (i) the fault pattern samples are more insufficient compared with the normal operation samples; (ii) the numbers of some fault pattern datasets are far less than that of other fault pattern datasets due to the different occurrence frequencies. To be specific, the primary reasons for inducing the imbalanced data of the chiller are analyzed as follows [41].

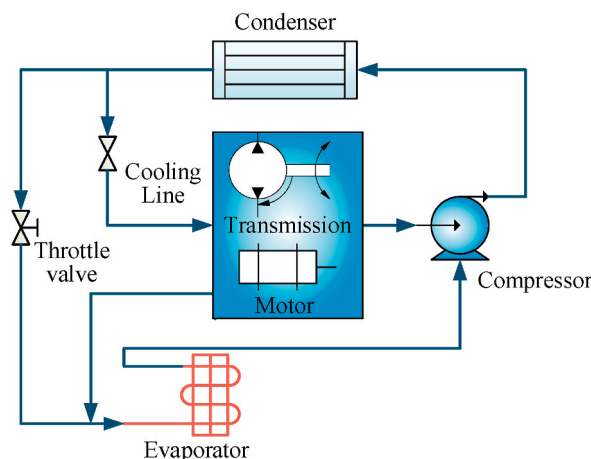


Fig. 1. The schematic diagram of the chiller [4,32].

- (1) During the running of the chiller, the duration of normal operation is much longer than that of the operation with faults. Even if a fault occurs in the chiller, the fault will be repaired timely to bring the chiller back to normal operation. The above two aspects make it difficult to collect the adequate fault pattern data samples.
- (2) In the real practical operation, if the rapid hard faults arise in the components of the chiller, such as the compressor, evaporator, and so on, will bring about the chiller shut down completely soon. Although the fault pattern data can be achieved by the experiments in the laboratory, the experiments not only need large economic costs but also give rise to potential danger. Therefore, the enough fault pattern data samples for the components of the chiller are not always available.
- (3) In the case of the faults happening, different fault patterns have different occurrence frequencies. Because of the unique operation characteristics of the chiller, some specific fault patterns possess much higher occurrence frequencies than that of other fault patterns. For instance, the occurrence probability of the reduced condenser water flow fault pattern is generally much larger than the occurrence probability of the refrigerant leak fault pattern.
- (4) For the fault patterns with different amplitudes, the probabilities of faults with large amplitudes are always lower than that of the faults with small amplitudes. Hence, the samples of the faults with large amplitude are more difficult to be measured and collected. For example, the samples of the Excess oil fault pattern with 50% amplitude are more difficult to be gathered than that of the Excess oil fault pattern with 14% amplitude, however, the former is more harmful to the chiller.
- (5) Before building the fault diagnosis models, the actually collected datasets are usually needed to discern the steady states and eliminate the outliers for the data preprocessing. However, the data preprocessing procedure is generally uncontrollable, which leads the balanced datasets to be imbalanced to a certain extent. The traditional chiller fault diagnosis models are constructed on the balanced fault pattern datasets by maximizing the overall diagnosis accuracy.

All in all, in the actual operation of the chiller, the observed and stored process datasets are often imbalanced due to multiple reasons. Nevertheless, most existing fault diagnosis models cannot be trained well on the imbalanced datasets to identify the fault patterns of the minority class snapshot samples. Therefore, according to the practical running of the chiller, it is very significant to investigate an improved chiller fault diagnosis approach under the imbalanced data conditions.

4. The developed HRT approach for handling the imbalanced data

In the practical operation, the historical fault pattern datasets of the chiller are always imbalanced due to the complex running characteristics. However, most traditional fault diagnosis models cannot be well trained on the imbalanced datasets of the chiller, which would result in much higher fault diagnosis error rates for the minority fault patterns. To figure out the problem of data imbalance, in our work, a new HRT method is constructed to rebalance the imbalanced historical fault pattern datasets of the chiller by combining the SMOTE over-sampling and the random under-sampling. In the beginning, the original imbalanced datasets of the chiller are supposed to involve the C kinds of the fault pattern. In the developed HRT method, one of the C classes imbalanced fault pattern subsets is determined as the benchmark dataset by the users. For the minority class datasets whose numbers of the samples are smaller than that of the benchmark dataset, the SMOTE over-sampling is applied to augment the samples to enlarge the minority datasets to be balanced. Besides, the artificially augmented samples have an identical clustering effect with the original minority datasets. For the majority class datasets whose numbers of the samples are larger than that of the benchmark dataset, the random under-sampling is adopted to randomly eliminate the samples to shrink the datasets. In this way, the imbalanced fault pattern datasets can be rebalanced to have the same number of samples as that of the benchmark dataset. The schematic diagram of hybrid sampling is shown in Fig. 2.

4.1. The SMOTE over-sampling method

The goal of the SMOTE over-sampling algorithm is to synthesize new samples through a linear combination of the minority samples and their neighboring samples, so as to expand the number of samples in the minority dataset [6,7]. The flowchart of the SMOTE over-sampling algorithm is displayed in Fig. 3.

Let the dataset X_0 containing n_0 data samples indicates the determined the benchmark dataset in the C classes imbalanced datasets, and the current resampling minority dataset is expressed as the $X_{\min}$ involving $n_{\min}$ data samples. The new artificial sample $x_{\text{new}}(j), j = 1, 2, \dots, n_0 - n_{\min}$ is produced by randomly combining the original sample x_i in the dataset $X_{\min}$ with its nearest neighbors. The nearest neighbor samples of the original sample x_i can be sought out by calculating the Euclidean distance $d(x_i, x_j)$ as follows.

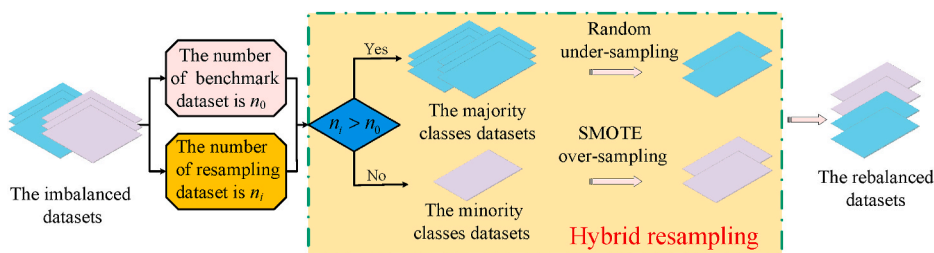


Fig. 2. The flowchart of the developed HRT.

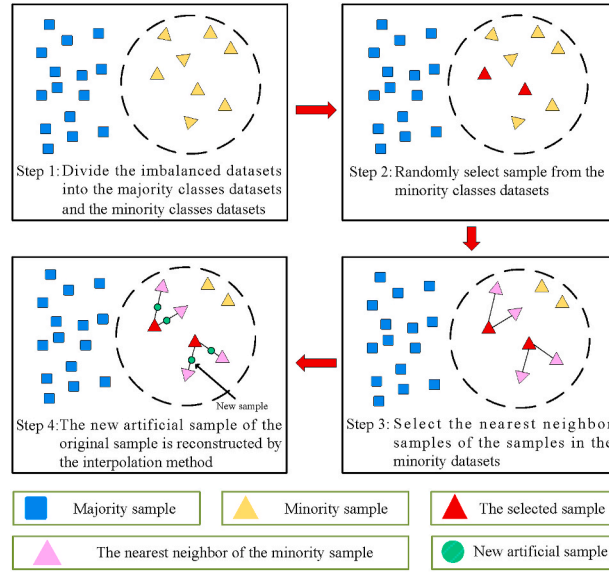


Fig. 3. The schematic diagram of the SMOTE over-sampling method.

$$d(x_i, x_f) = \sqrt{\|x_i - x_f\|^2}, f = 1, 2, \dots, n_{\min} - 1 \quad (1)$$

where x_f represents the sample in the dataset $X_{\min}$ except for the sample x_i .

After the Euclidean distances between the sample x_i and the samples $x_f, f = 1, 2, \dots, n_{\min} - 1$ are computed, these Euclidean distances are sorted from small to large. In this manner, the distance vector D constructed utilizing the rearranged Euclidean distances is expressed as

$$D = [d(x_i, x_1), d(x_i, x_2), \dots, d(x_i, x_k), \dots, d(x_i, x_{n_{\min}-1})]^T \quad (2)$$

where x_1 indicates the sample in the minority dataset $X_{\min}$ corresponding to the smallest Euclidean distance $d(x_i, x_1)$, x_2 denotes the sample in the dataset $X_{\min}$ with the second smallest Euclidean distance $d(x_i, x_2)$, and so on.

The samples $x_k, k = 1, 2, \dots, K$ corresponding to the first K smallest Euclidean distances $d(x_i, x_1), d(x_i, x_2), \dots, d(x_i, x_K)$ are chosen as the K nearest neighbor samples of the sample x_i . Subsequently, the mean value $\bar{x}_k$ of the K nearest neighbors is calculated as

$$\bar{x}_k = \frac{1}{K} \sum_{k=1}^K x_k \quad (3)$$

After that, the new artificial sample $x_{\text{new}}(j)$ of the original sample x_i is reconstructed as

$$x_{\text{new}}(j) = x_i + \text{rand}(0, 1) \cdot |x_i - \bar{x}_k| \quad (4)$$

where the symbol $\text{rand}(0, 1)$ indicates random distribution on the interval of $[0, 1]$.

Note that, in the case of the SMOTE over-sampling algorithm, the relationship between the benchmark dataset samples n_0 and the current minority dataset samples $n_{\min}$ can be expressed as

$$n_0 = a \cdot n_{\min} + b \quad (5)$$

where $a \neq 0$ and $b < n_{\min}$.

Based on Eq. (5), there are two cases to randomly synthesize $n_0 - n_{\min}$ samples, which are formulated as follows.

- (1) Under the circumstance of $b = 0$, all the $n_{\min}$ samples in the current minority dataset are adopted to randomly reconstruct the $n_0 - n_{\min}$ samples by repeating the above introduced procedure of the SMOTE over-sampling.
- (2) In the case of $b \neq 0$, the original b samples are first stochastically selected from the current minority dataset. Then, the selected original b samples are utilized to produce the new artificial synthesized b samples and the $n_{\min}$ current minority dataset samples are finally employed to randomly generate the $n_{\min} \cdot (a - 1)$ by repeating the procedure of the SMOTE over-sampling.

4.2. The random under-sampling algorithm

When the number of samples $n_{\max}$ in the current resampling dataset $X_{\max}$ is larger than the samples n_0 in the benchmark dataset X_0 , i.e., $n_{\max} > n_0$, the random under-sampling algorithm [8,9] exhibited in Fig. 4 is carried out to randomly eliminate some samples in the

current majority class dataset $X_{\max}$.

All the $n_{\max}$ samples in the current majority dataset $X_{\max} = [x_{\max}(1), x_{\max}(2), \dots, x_{\max}(n_{\max})]^T$ are first stochastically rearranged to establish a new dataset $X_{\max}^{\text{new}}$.

$$X_{\max}^{\text{new}} = [x_{\max}(r(1)), x_{\max}(r(2)), \dots, x_{\max}(r(n_{\max}))]^T \quad (6)$$

where the vector $x_{\max}(i), i = 1, 2, \dots, n_{\max}$ represents the i -th sample in the majority class dataset $X_{\max}$. The vector $r = [r(1), r(2), \dots, r(n_{\max})]^T$ denotes a random vector whose elements $r(i), i = 1, 2, \dots, n_{\max}$ are the randomly selected integers form the interval of $[1, n_{\max}]$ without repetition.

What's more, the random vector r is defined as follows.

$$r = \text{randperm}(n_{\max}, n_{\max}) \quad (7)$$

where the $\text{randperm}(n_{\max}, n_{\max})$ means to generate the $n_{\max}$ random integers form 1 to $n_{\max}$ without repetition.

Then, the first n_0 samples in the dataset $X_{\max}^{\text{new}}$ remained to construct the rebalanced dataset $X_{\max}^{\text{reb}}$, while the other superfluous samples are removed. In particular, the rebalanced dataset $X_{\max}^{\text{reb}}$ is reconstructed as

$$X_{\max}^{\text{reb}} = [x_{\max}(r(1)), x_{\max}(r(2)), \dots, x_{\max}(r(n_0))]^T \quad (8)$$

4.3. The implementation of the developed HRT technique

As described in Fig. 5, the detailed procedure of the proposed HRT approach is given out in our work. The HRT is indeed an integration of the SMOTE over-sampling and the random under-sampling, which has the ability to effectively settle the problem of the imbalanced fault pattern datasets. To be specific, the benchmark dataset X_0 with the number of samples n_0 is first selected from the C classes imbalanced fault pattern datasets. Afterward, the number of samples n_i in the current i -th resampling dataset X_i is compared with the number of samples n_0 . If n_i is smaller than n_0 , the current resampling dataset X_i is regarded as the minority class dataset $X_{\min}$, and the SMOTE over-sampling is subsequently utilized to generate the $n_0 - n_{\min}$ artificially synthesized samples. If n_i is larger than n_0 , the current resampling dataset X_i is identified as the majority class dataset $X_{\max}$, and the random under-sampling is then employed to randomly wipe off the $n_{\max} - n_0$ samples. The above procedures are repeated $C - 1$ times to rebalance the C kinds of the imbalanced fault pattern datasets.

5. The proposed ELM diagnosis model based on an improved selective ensemble learning strategy

After the imbalanced fault pattern datasets are rebalanced, the ELM network is adopted to diagnose the faults of the chiller on account of the ELM's fast learning speed and good generalization ability. Whereas, because the fault diagnosis accuracy and robustness of the single ELM model are unsatisfied, in our work, the number of C various ELM basic models are built on the C diverse rebalanced training datasets. Specifically, every fault pattern dataset in the C classes imbalanced datasets is determined as the benchmark dataset X_0 , hence the developed HRT method is repeated C times to achieve a total of C rebalanced training datasets. It should be noted that each rebalanced training dataset still includes the C kinds of fault pattern subsets. To effectively choose the basic ELM models owning the superior fault diagnosis performances from the number of C basic ELM models, a revised ensemble learning ELM based scheme is established with the aid of putting forward a novel selective ensemble learning strategy in this paper. At last, the selected multiple basic ELM models are integrated together to construct the final ensemble diagnosis model, for the purpose of identifying the pattern of the fault snapshot dataset. In addition, the improved selective ensemble learning based ELM diagnosis strategy also contributes to reducing the computation complexity of the final ensemble model. The detailed schematic diagram of the proposed selective ensemble learning based ELM diagnosis strategy is displayed in Fig. 6.

5.1. The multiple basic ELM models training

In this paper, the developed HRT approach is implemented repetitively to resample a total of C rebalanced training datasets by

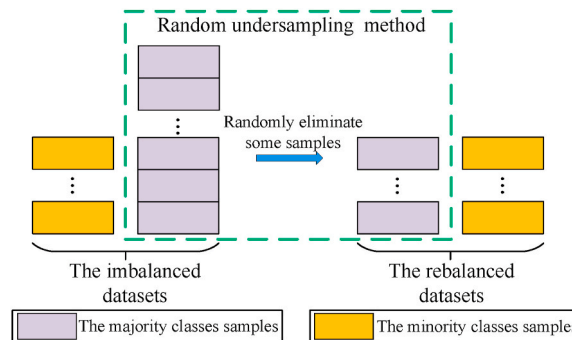


Fig. 4. The flowchart of the random under-sampling algorithm.

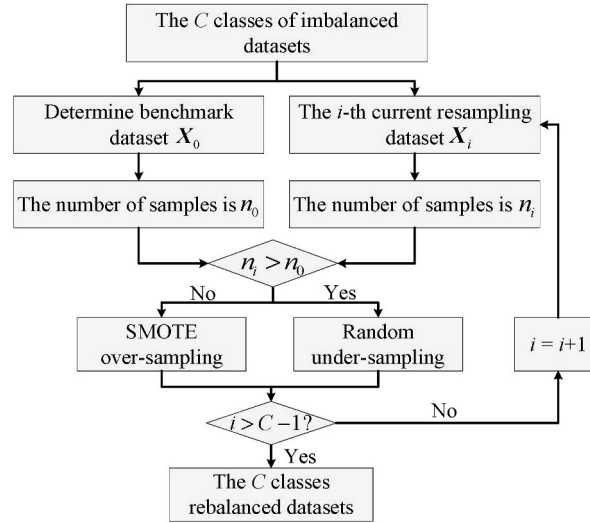


Fig. 5. The detailed procedure of the proposed HRT approach.

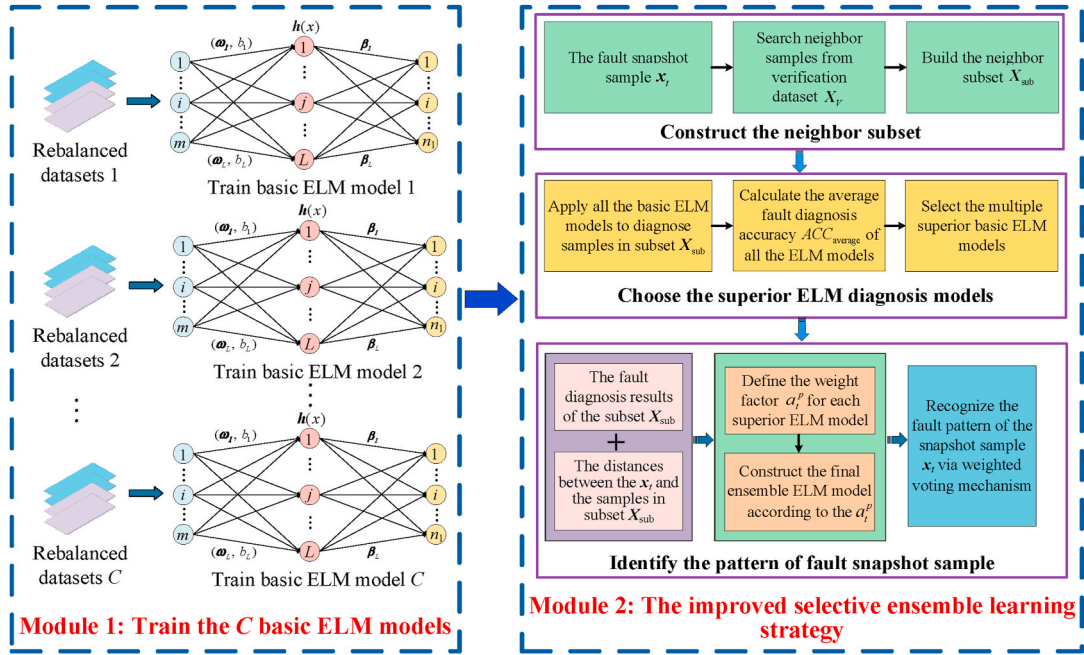


Fig. 6. The schematic diagram of suggested selective ensemble learning based ELM diagnosis strategy.

regarding every dataset in the C kinds of imbalanced fault pattern datasets as the benchmark dataset. Subsequently, as illustrated in Fig. 7, the number of C various basic ELM models are trained on these rebalanced training datasets to manage the drawbacks of single ELM model's unsatisfactory fault diagnosis results and robustness. Based on a specific rebalanced training dataset, one of the C various basic ELM fault diagnosis models is built as follows.

As shown in Fig. 8, the ELM is a special single hidden layer feed-forward neural network (SLFNs) with an input layer, output layer, and hidden layer, which can provide good classification performance at extremely fast training speed [42].

For the specific rebalanced training dataset $\{X, Y\} = \{x_i, y_i\}_{i=1}^N$ involving the C kinds of fault pattern subsets, where N is the number of the whole samples, x_i indicates a m -dimensional data sample and y_i represents a category attribute vector. The relationship between the weight vector β_i connecting the i -th hidden node to output nodes, the ELM's output o_j and the activation function $G(w_i \cdot x + b_i)$ is expressed as

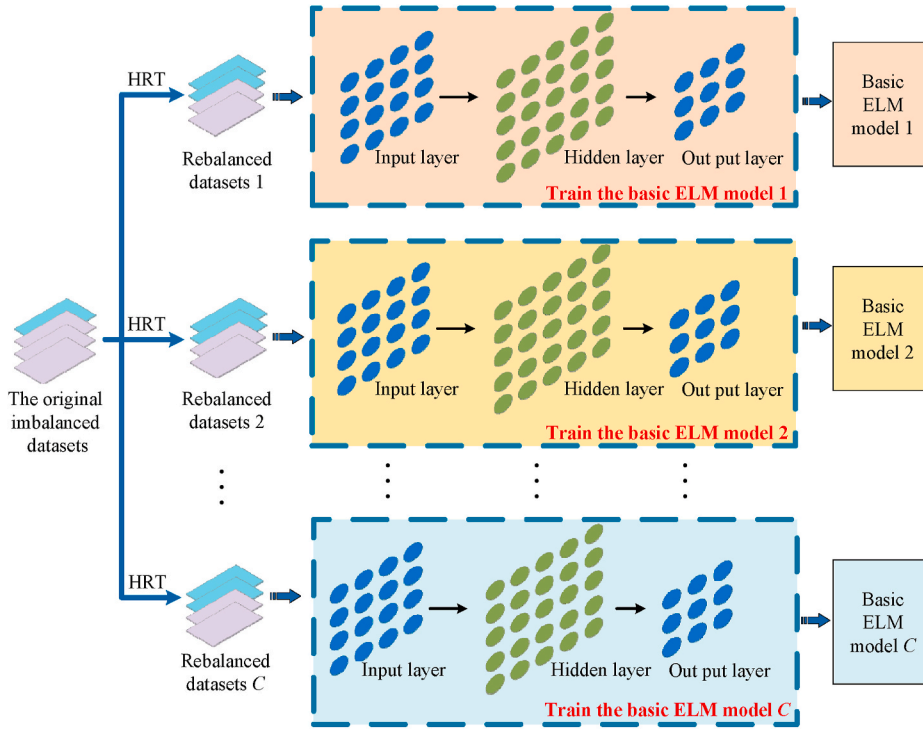


Fig. 7. The flowchart of training the multiple basic ELM fault diagnosis models.

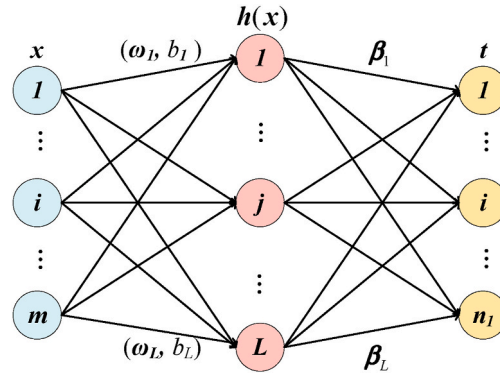


Fig. 8. The structure diagram of the basic ELM model.

$$\sum_{i=1}^L \beta_i G(\mathbf{w}_i \cdot \mathbf{x}_j + b_i) = o_j, j = 1, \dots, N \quad (9)$$

where L denotes the number of nodes in the hidden layer, $\mathbf{w}_i$ indicates the weight vector connecting input nodes to the i -th hidden node, and b_i indicates the bias of the i -th hidden node.

A notable feature of the ELM is that the weights $\mathbf{w}_i$ and the bias b_i could be obtained randomly in the light of continuous probability distribution. Particularly, the activation function $G(\mathbf{w}_i \cdot \mathbf{x} + b_i)$ is selected as the sigmoid function in this paper, which is formulated as

$$G(\mathbf{w}_i \cdot \mathbf{x} + b_i) = \frac{1}{1 + \exp(-(\mathbf{w}_i \cdot \mathbf{x} + b_i))} \quad (10)$$

To obtain much better classification effectiveness, the ELM is further enhanced to acquire both the smallest training error $\|\mathbf{O} - \mathbf{Y}\|^2$ and the smallest output weights norm $\|\beta\|^2$. The $\mathbf{O} = [o_1, o_2, \dots, o_N]^T \in R^{N \times n_1}$ indicates the output of the ELM, $\mathbf{Y} = [y_1, y_2, \dots, y_N]^T \in R^{N \times n_1}$ represents the target matrix of the rebalanced training dataset $\mathbf{X}$ and n_1 denotes the dimension of the output layer.

Therefore, the optimization objective of ELM is rewritten as [43].

$$\|\beta\|^2 + \frac{\tilde{C}}{2} \sum_{i=1}^N \|e_i\|^2$$

$$s. t. \mathbf{h}(\mathbf{x}_i)\beta = \mathbf{y}_i^T - \mathbf{e}_i^T, i = 1, 2, \dots, N$$
(11)

where $\mathbf{e}_i$ represents the error vector of the output nodes related to the i -th data sample $\mathbf{x}_i$, and $\tilde{C}$ indicates a penalty coefficient associated with errors.

From the objective function of the ELM, the smaller the training error and the weight norm are, the better the generalization performance of the ELM tends to possess. Therefore, the ELM's objective function provides a solid theoretical foundation for good generalization performance.

According to the theory of Karush-Kuhn-Tucker (KKT), the objective function of the ELM is equivalent to the dual optimization problem of the following equation.

$$L_{\text{ELM}} = \min_{\beta, \alpha_i} \frac{1}{2} \|\beta\|^2 + \frac{\tilde{C}}{2} \sum_{i=1}^N \|e_i\|^2 - \sum_{i=1}^N \alpha_i (\mathbf{h}(\mathbf{x}_i)\beta - \mathbf{y}_i^T + \mathbf{e}_i^T)$$
(12)

where α_i indicates the Lagrange multiplier.

The output weight vector β can be obtained by solving the above optimization problem. After the vector β is determined, the ELM's output matrix $\mathbf{O} = [\mathbf{o}_1, \mathbf{o}_2, \dots, \mathbf{o}_N]^T$ of the rebalanced training dataset $\mathbf{X}$ can be calculated as

$$\mathbf{O} = \mathbf{H}\beta$$
(13)

where $\mathbf{H}$ represents the mapping matrix that randomly transforms the original rebalanced training dataset $\mathbf{X}$ into high-dimensional nonlinear space.

$$\mathbf{H} = \begin{bmatrix} \mathbf{h}(\mathbf{x}_1) \\ \vdots \\ \mathbf{h}(\mathbf{x}_N) \end{bmatrix} = \begin{bmatrix} G(\mathbf{w}_1 \cdot \mathbf{x}_1 + b_1) & \cdots & G(\mathbf{w}_L \cdot \mathbf{x}_1 + b_L) \\ \vdots & \cdots & \vdots \\ G(\mathbf{w}_1 \cdot \mathbf{x}_N + b_1) & \cdots & G(\mathbf{w}_L \cdot \mathbf{x}_N + b_L) \end{bmatrix}_{N \times L}$$
(14)

In our work, we repeat the above recommended training procedure of the ELM on the number of C diverse rebalanced training datasets to gain the C various basic ELM models.

5.2. The improved selective ensemble learning strategy

Due to the discrepant fault diagnosis performance of the C basic ELM models founded on the various rebalanced training datasets, we come up with an improved selective ensemble learning strategy to pick out the basic ELM models with superior diagnosis capabilities to set up a novel ensemble ELM based fault diagnosis scheme for the chiller. As exhibited in Fig. 9, the k nearest neighbor samples of the current fault snapshot sample $\mathbf{x}_i$ are searched in the verification dataset X_V which also contains the C classes fault pattern subsets, then all the C basic ELM models are applied to diagnose the fault patterns of the determined k nearest neighbors related

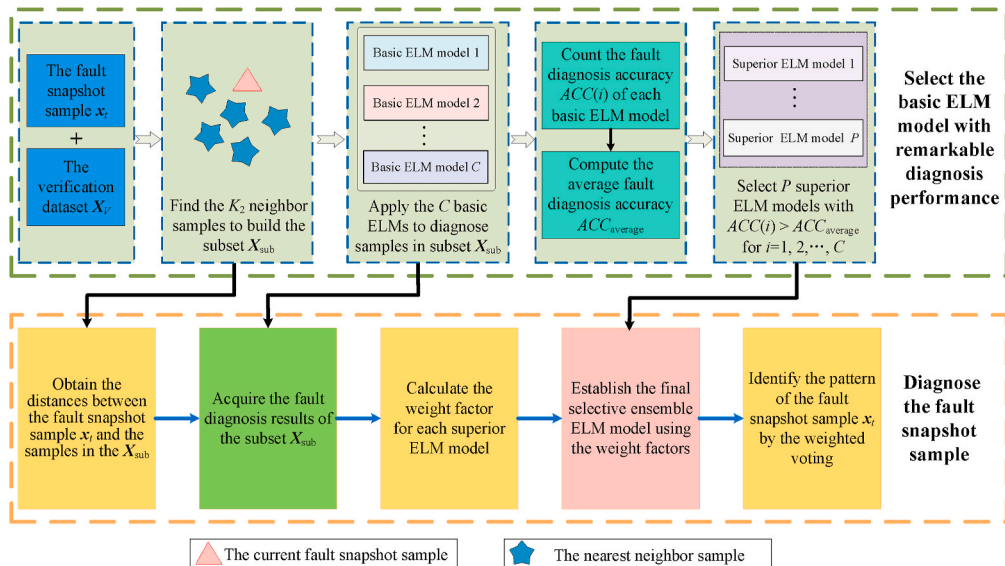


Fig. 9. The schematic diagram of the improved selective ensemble learning strategy.

to the x_t . Afterward, the basic ELM fault diagnosis models whose diagnosis accuracies are greater than the average accuracy of all the basic ELM models are chosen and adopted to construct the final ensemble ELM model according to the weight factor defined in our work. The constructed ensemble ELM based model is eventually employed to recognize the pattern of the fault snapshot dataset by virtue of the weighted voting mechanism, which reveals the remarkable improvement of the fault diagnosis effectiveness as well as the significant reduction of the computation complexity.

5.2.1. Select the basic ELM model with remarkable diagnosis performance

Given the number of C basic ELM fault diagnosis models and the validation dataset X_V consisting of the C kinds of fault patterns, the specific implementation steps of the proposed selective ensemble learning strategy are revealed in Fig. 10.

For the current fault snapshot sample x_t , the Euclidean distances between the sample x_t and all the samples in the verification dataset $X_V = [x_v(1), x_v(2), \dots, x_v(N_V)]^T$ are calculated as

$$d(x_t, x_v(i)) = \sqrt{\|x_t - x_v(i)\|^2}, i = 1, 2, \dots, N_V \quad (15)$$

where N_V represents the number of samples in the verification dataset X_V .

After the N_V Euclidean distances $d(x_t, x_v(1)), d(x_t, x_v(2)), \dots, d(x_t, x_v(N_V))$ are obtained, these Euclidean distances are sorted from small to large. The samples $x_v(k), k = 1, 2, \dots, K_2$ in the dataset X_V corresponding to the first K_2 smallest Euclidean distances are selected to establish the nearest neighbor sample subset X_{sub} . Note that the fault pattern of these K_2 samples in the verification dataset X_V is known in advance.

In accordance with the k -nearest neighbor algorithm [43], the current fault snapshot sample x_t can be represented by the weight of its K_2 nearest neighbors in the established subset X_{sub} . Consequently, the fault diagnosis result of the sample x_t is equivalent to the fault pattern identification results of its K_2 neighbors in the neighbor subset X_{sub} .

Based on the above principle, all the C various basic ELM fault diagnosis models are utilized to recognize the fault pattern of the K_2 nearest neighbors in the subset X_{sub} . After that, the average fault diagnosis accuracy $ACC_{average}$ of the C basic ELM models is computed as

$$ACC_{average} = \frac{1}{C} \sum_{i=1}^C ACC(i) \quad (16)$$

where the $ACC(i), i = 1, 2, \dots, C$ denotes the fault diagnosis accuracy of the neighbors in the subset X_{sub} for the i -th basic ELM model.

The P basic ELM model whose fault diagnosis accuracy $ACC(p), p = 1, 2, \dots, P$ is larger than the average accuracy $ACC_{average}$ is selected to constitute the final ensemble ELM fault diagnosis model.

5.2.2. Identify the fault pattern of the snapshot sample

As is known to all, the closer distances the data samples reveal, the higher similarities these data samples have. As a consequence, we determine the weights of the P selected basic ELM fault diagnosis models by jointly considering the fault pattern identification

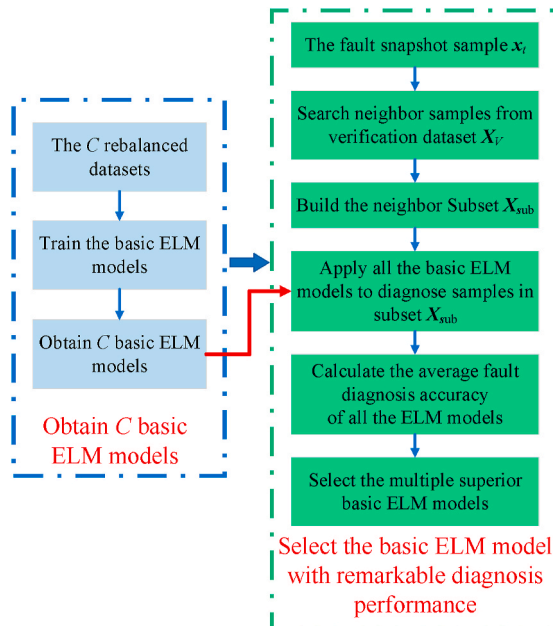


Fig. 10. The procedure of the superior basic ELM fault diagnosis models selection strategy.

results of the samples in the neighbor subset X_{sub} as well as the distances between the fault snapshot sample x_t and its K_2 nearest neighbors. Specifically, to integrate the final ensemble ELM based fault diagnosis model, the weight factor α_t^p of the p -th chosen basic ELM model is calculated in our work as follows:

$$\alpha_t^p = \frac{1}{D_t} \sum_{i=1}^{K_2} \frac{I(o_v^i, y_v^i)}{d(x_t, x_v(i))} \quad (17)$$

where $d(x_t, x_v(i))$ denotes the Euclidean distance between the fault snapshot sample x_t and its i -th nearest neighbors $x_v(i)$. $D_t = \sum_{p=1}^P \alpha_t^p$ indicates the normalized factor, y_v^i represents the real category label vector of the i -th nearest neighbor $x_v(i)$ and o_v^i is the output of the p -th basic ELM model corresponding to the sample $x_v(i)$.

In addition, $I(o_v^i, y_v^i)$ is determined as follows in this paper.

$$I(o_v^i, y_v^i) = \begin{cases} 1 & , \text{ if } \operatorname{argmax}(o_v^i(j)) = \operatorname{argmax}(y_v^i(j)) \quad , j \in \{1, \dots, C\} \\ 0 & , \text{ else} \end{cases} \quad (18)$$

where $\operatorname{argmax}(y_v^i(j))$ represent the real category of the neighbor sample $x_v(i)$ and $\operatorname{argmax}(o_v^i(j))$ indicate the p -th basic ELM's fault pattern recognition result related to the neighbor sample $x_v(i)$.

At last, the pattern of the fault snapshot sample x_t is identified by the finally constructed selective ensemble ELM based fault diagnosis model according to a weighted voting mechanism. Particularly, let $o_t^p(i)$ denote the value of the i -th output node in the p -th selected basic ELM model when classifying the sample x_t , the category label of the fault snapshot sample x_t is determined as

$$\text{label}(x_t) = \operatorname{argmax} \sum_{p=1}^P \alpha_t^p o_t^p(i), i \in \{1, \dots, C\} \quad (19)$$

6. The presented HRIELM based chiller fault diagnosis in the imbalanced data environment

As illustrated in Fig. 11, our proposed HRIELM based chiller fault diagnosis scheme involves two portions: the HRT based multiple basic ELMs offline modeling stage and the pattern of fault snapshot sample online diagnosis stage. Specifically, the HRT based multiple basic ELMs offline modeling stage consists of the hybrid resampling module and the basic ELM models training module, while the pattern of fault snapshot sample online diagnosis stage contains the selective ensemble ELM model construction module and the fault pattern identification module. The details of the developed HRIELM based fault diagnosis strategy are expressed as follows.

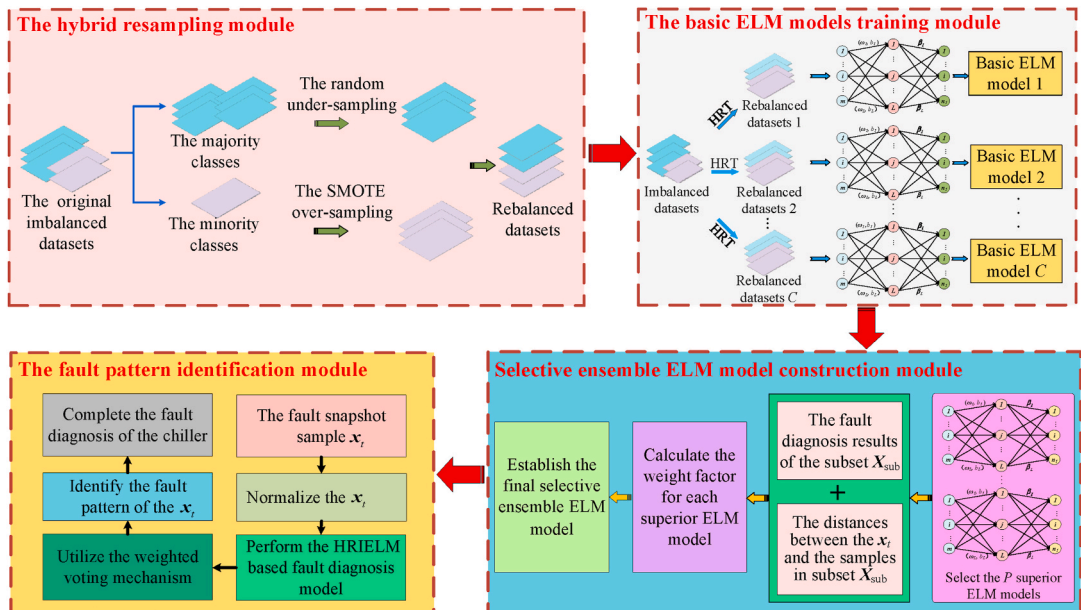


Fig. 11. The flowchart of the suggested HRIELM based fault identification scheme.

6.1. The HRT based multiple basic ELMs offline modeling

- (1) Collect the C kinds of historical fault pattern datasets to construct the original imbalanced dataset X_{imb} and the validation dataset X_v . The two datasets X_{imb} and X_v are then normalized to be zero mean and unit variance according to the original imbalanced dataset X_{imb} .
- (2) Regard the i -th fault pattern dataset X_{F_i} , $i = 1, 2, \dots, C$ as the benchmark dataset X_0 . The proposed HRT approach is adopted to resample the original imbalanced dataset X_{imb} to achieve a rebalanced training dataset $X_{reb}(i)$.
- (3) Repeat the HRT approach C times to obtain a total of C different rebalanced training datasets $X_{reb}(1), X_{reb}(2), \dots, X_{reb}(C)$, by treating each of the fault pattern datasets $X_{F_1}, X_{F_2}, \dots, X_{F_C}$ as the benchmark dataset.
- (4) Train the number of C various basic ELM fault diagnosis models on these diverse rebalanced training datasets $X_{reb}(1), X_{reb}(2), \dots, X_{reb}(C)$ via implementing the training procedure of the ELM recommended in Section 5.1 C times.

6.2. The pattern of fault snapshot sample online diagnosis

- (1) Establish the neighbor subset X_{sub} by searching the nearest neighbor samples of the fault snapshot sample x_t in the verification dataset X_v .
- (2) Employ all the C various trained basic ELM models to recognize the fault pattern of the samples in the neighbor subset X_{sub} . The average fault diagnosis accuracy $ACC_{average}$ is then calculated according to Eq. (16).
- (3) Select the P superior basic ELM models whose fault diagnosis accuracies $ACC(p)$, $p = 1, 2, \dots, P$ for the neighbor subset X_{sub} are greater than the average accuracy $ACC_{average}$.
- (4) Determine the weights of the P selected basic ELM models based on Eq. (17) to integrate the final ensemble ELM based fault diagnosis model. The pattern of the fault snapshot sample x_t is identified by the finally constructed selective ensemble ELM according to a weighted voting mechanism defined in Eq. (19).

7. The experiments and comparisons on the chiller with imbalanced data

7.1. Introduction of the experimental imbalanced data

In this paper, the American Society of Heating, Refrigerating, and Air-conditioning Engineers (ASHRAE) research project No. RP-1043 dataset available at <https://www.techstreet.com/ashrae/searches/36470950> is utilized to prove the fault diagnosis capability of the developed HRIELM method in detail. The RP-1043 experimental dataset is a benchmark dataset, which is commonly adopted to examine the effectiveness of the fault identification approaches [32,40,44,45]. In order to reveal the influences of different fault severity levels on the fault diagnosis performance, the original dataset is further segmented into 4 diverse data subsets, i.e., Level-1, Level-2, Level-3, and Level-4, which represent the 4 different severity levels. The fault severity levels in these four subsets change from small to large, and each subset contains seven historical fault pattern datasets and one normal operation dataset. Detailed information on seven classic fault patterns under different severity levels and their abbreviations are shown in Table 1. According to references [32, 44], 16 process variables are applied to perform the fault diagnosis of the chiller.

In our work, each fault severity level dataset consists of seven different fault pattern subsets shown in Table 1. To relieve the computation complexity, each fault pattern subset in the different severity level datasets is set to contain 1000 fault samples. For each fault severity level, 200 fault samples are randomly selected from each fault pattern subset to construct the corresponding fault snapshot dataset, and 80 samples are stochastically chosen from each fault pattern subset to establish the validation dataset. Therefore, there are respectively a total of 1400 fault samples and 560 fault samples in the whole seven fault snapshot datasets and the validation dataset for each fault severity level.

Moreover, to generate the original imbalanced dataset for each fault severity level, seven various historical fault pattern subsets with different sample numbers are produced from the remaining fault samples according to the data imbalance rate. Specifically, the minimum sample number of the minority historical fault pattern subset and the maximum sample number of the minority historical fault pattern subset are first calculated via a given data imbalance rate. Afterward, the sample numbers of other five imbalanced historical fault pattern subsets are stochastically determined within the range of the minimum and maximum sample numbers. Hence, in this paper, the fault sample numbers of the seven imbalanced historical fault pattern subsets in the original imbalanced dataset are changing along with the value of the data imbalance rate.

Table 1
Different fault patterns and four different fault severity levels.

Fault patterns	Four fault severity levels			
	Level-1 (%)	Level-2 (%)	Level-3 (%)	Level-4 (%)
Condenser fouling (CF)	12	20	30	45
Excess oil (EO)	14	32	50	68
Refrigerant leak (RL)	−10	−20	−30	−40
Refrigerant overcharge (RO)	10	20	30	40
Reduced condenser water flow (FWC)	−10	−20	−30	−40
Reduced evaporator water flow (FWE)	−10	−20	−30	−40
Non-condensable gas in refrigerant (NC)	1.0	1.7	2.4	5.7

7.2. Comparative methods and parameters setting

For the sake of testifying the superiority of the developed HRIELM based fault diagnosis method, the bagging ELM (BGELM), boosting ELM (BTELM), weighted ELM (WELM), and boost WELM (BWELM) based methods are contrasted with the suggested HRIELM based approach. In the HRT based resampling method, the number of the nearest neighbor samples in the original minority classes is selected as five. The activation function of the HRIELM is chosen as sigmoid function according to references [12,13]. The number of HRIELM's hidden neurons and its trade-off parameter are respectively selected from the ranges {1000, 2000, ..., 9000} and

$\{10^{-5}, 10^{-4}, \dots, 10^5\}$ by the grid search according to the fault diagnosis performance. In our work, the number of hidden neurons and the trade-off parameter of the HRIELM are finally chosen as 2000 and 0.01, respectively. The number of basic ELM models is determined according to the number of fault pattern datasets in the original imbalanced dataset, thus seven basic ELM models are established for the subsequent presented selective ensemble learning strategy. Because 16 process variables are employed to carry out the fault diagnosis and the fault snapshot sample can be classified into seven fault patterns, hence 16 neurons and 7 neurons are respectively determined in each ELM's input layer and output layer. For each fault snapshot sample, its five nearest neighbor samples are searched in the verification dataset to build the nearest neighbor subset.

To make a fair comparison, the sigmoid function is adopted as the activation function and the number of hidden neurons is set to be 2000 in the BGELM, BTELM, WELM, and BWELM based approaches. The values of the trade-off parameter for the BGELM, BTELM, and BWELM are all set as 1, while the value of the trade-off parameter for the WELM method is chosen as 10 empirically. In addition, in the BTELM and BWELM, seven ELM models are trained as the basic ELM models, and the rate of training subset to the original train dataset is set as 0.9. When the misclassification rate of the basic ELM models is less than 0.5, the weight of the basic ELM model is calculated and the sample distribution is updated at the same time; otherwise, the current ELM basic model will be abandoned. Besides, the maximum number of training times T is set as 50. In the BGELM, the number of basic ELM models is seven, and the rate of training subset to the original train dataset is set as 0.8.

7.3. The fault diagnosis evaluation indices

In order to appraise the diagnosis performance of the proposed HRIELM method under the imbalanced data environments, the data imbalance rate (IR) index is introduced to measure the imbalance degree of the training datasets, which is expressed as

$$IR = \frac{\min(N_i)}{\max(N_i)}, i = 1, 2, \dots, C \quad (20)$$

where $N_i, i = 1, 2, \dots, C$ denotes the number of samples in each fault pattern dataset, $\min(N_i)$ indicates the smallest number of samples in the minority fault pattern dataset, $\max(N_i)$ represents the largest number of samples in the majority fault pattern dataset. Note that, for the original imbalanced chiller datasets with different values of the index IR , the minority and majority fault pattern datasets are randomly determined instead of remaining constant.

To assess the fault identification ability of the proposed HRIELM and other four compared methods, three different fault diagnosis accuracy indices, i.e., the fault diagnosis accuracy $ACC(i), i = 1, 2, \dots, C$ of the fault snapshot samples in the i -th fault pattern dataset, the overall fault diagnosis accuracy ACC_{total} of all the fault snapshot samples in the C fault pattern datasets and the geometric mean of the fault diagnosis accuracy ($G-mean$), are employed.

Particularly, the index $ACC(i)$ is defined as

$$ACC(i) = \frac{N_i(i)}{N_i}, i = 1, \dots, C \quad (21)$$

where $N_i(i)$ represents the number of correctly identified fault snapshot samples in the i -th fault pattern dataset.

The index ACC_{total} indicates the ratio of the number of all the correctly identified fault snapshot samples to the total number of the fault snapshot samples in the C fault pattern datasets, which is calculated as

$$ACC_{total} = \frac{\sum_{i=1}^C N_i(i)}{\sum_{i=1}^C N_i} \quad (22)$$

Considering the data imbalance problem of the chiller, the geometric mean of the fault diagnosis accuracy, i.e., the $G-mean$, for every fault snapshot dataset computed in Eq. (23) is further utilized to estimate the fault diagnosis effectiveness. The index $G-mean$ is

Table 2

Sample numbers in the imbalanced and rebalanced datasets for $IR = 0.05$.

Fault pattern	Sample number in the imbalanced dataset	Sample number in the rebalanced dataset
CF	63	350
EO	700	350
FWC	350	350
FEW	246	350
NC	36	350
RL	381	350
RO	367	350

one of the most commonly used indices to evaluate the fault identification performance under imbalanced data environments [46].

$$G - mean = \left(\prod_{i=1}^c ACC(i) \right)^{1/c} \quad (23)$$

7.4. Experimental results and comparisons

7.4.1. The performance analysis of the developed HRT approach

In order to testify the resampling performance of our developed HRT method when handling the imbalanced datasets of the chiller, the original imbalanced dataset $IR = 0.05$ corresponding to the fault severity degree Level-1 is first adopted in our work. As previously mentioned, the fault patterns in the fault severity degree Level-1 can be considered as incipient faults because they have the smallest severity degree. Table 2 gives out the different numbers of the samples in the seven diverse fault pattern datasets. As displayed in Table 2, the dataset of fault pattern EO has the largest number of samples, which involves 700 samples. While the dataset of fault pattern NC with only 36 samples owns the smallest number of samples. Therefore, these original fault pattern datasets are extremely imbalanced. The dataset of fault pattern FWC containing 350 samples is regarded as the current benchmark dataset, and the HRT method is carried out only once to rebalance these imbalanced fault pattern datasets. From Table 2, after implementing the HRT method, the seven different fault pattern datasets are all resampled to be balanced datasets, each of which possesses 350 samples. To make a more intuitionistic comparison, the number of the samples in these seven different fault pattern datasets before and after performing the presented HRT approach are illustrated in Fig. 12.

To further confirm the superior effectiveness of the suggested HRT method for rebalancing the imbalanced datasets, the HRT resampling approach is compared with an adaptive synthetic (ADASYN) sampling approach [47,48] which is a typical over-sampling method in this section. Three imbalanced datasets of the fault patterns EO, NC, and FWC shown in Table 2 before and after rebalancing are visualized in the low-dimension space by taking advantage of the principal component analysis (PCA) technique. For the HRT resampling approach, the chosen three datasets of the fault patterns EO, NC, and FWC respectively represent the majority class, minority class, and the benchmark class datasets. Considering the ADASYN is an over-sampling method, the fault pattern EO dataset containing the largest number of samples is regarded as the benchmark class dataset and the fault patterns NC and FWC are both determined as the minority class datasets during the implementation of the ADASYN based data rebalancing algorithm. Fig. 13 reveals the data separation results of these three fault pattern datasets before and after resampling by the HRT and the ADASYN in the PCA principal component space. According to Fig. 13 (a), the three imbalanced fault pattern datasets before performing the HRT and the ADASYN are clearly separated from each other. From Fig. 13(b), we can see that the three rebalanced fault pattern datasets after performing the HRT are also projected far away from each other. Besides, these three rebalanced datasets are evidently overlapped together with the corresponding imbalanced datasets. As shown in Fig. 13 (c), although the rebalanced fault pattern FWC dataset is also overlapped together with the corresponding imbalanced datasets after implementing the ADASYN, many synthetic samples in the rebalanced fault pattern NC dataset clearly deviate from the original NC dataset. This phenomenon exhibits that the data rebalancing effect of the ADASYN is worse than that of the developed HRT. The scatter diagrams of these three fault pattern datasets before and after carrying out the HRT and the ADASYN, which achieved in the PCA residual space, are illustrated in Fig. 14. As can be seen from Fig. 14 (b), for the HRT, data separation and overlapping results of these three imbalanced and rebalanced fault pattern datasets are similar to the scatter diagrams exhibited in Fig. 13. However, for the ADASYN shown in Fig. 14 (c), most synthetic samples of the rebalanced fault patterns NC and FWC datasets are clearly separated from the corresponding imbalanced datasets, while only a small amount of the synthetic samples are overlapped together with the previous ones. This again manifests that the HRT outperforms the ADASYN in terms of rebalancing the imbalanced datasets.

Based on the experimental and comparative results given above, we can see that the developed HRT approach can not only balance the numbers of samples in the various fault pattern datasets but also keep the resampled samples having strong consistency with the samples in original datasets. As a consequence, the developed HRT approach is very suitable for solving the problem of the chiller's data imbalance, which is significantly beneficial to the improvement of the chiller fault diagnosis capability.

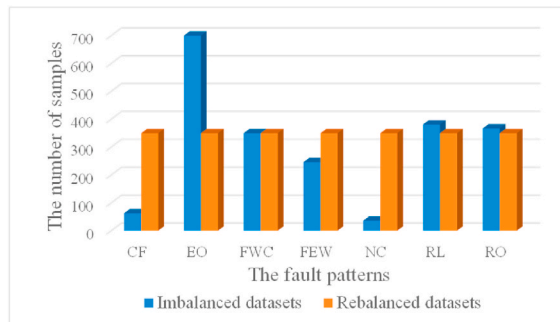
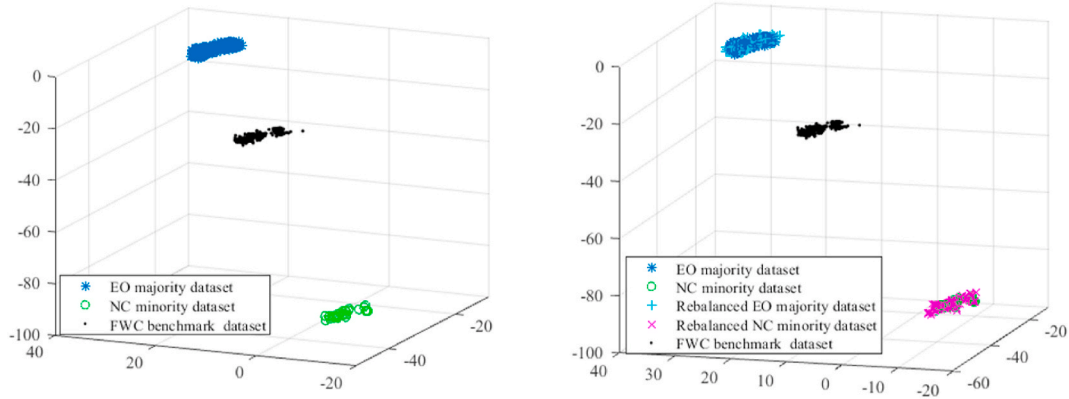
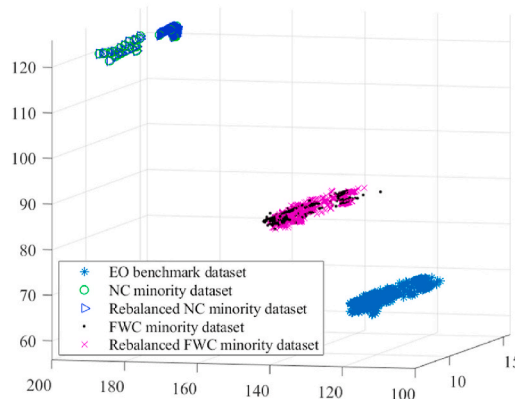


Fig. 12. The histogram of the sample numbers in the imbalanced and rebalanced datasets for $IR = 0.05$.



(a) The original imbalanced datasets.

(b) The imbalanced datasets and rebalanced datasets by the HRT.



(c) The imbalanced datasets and rebalanced datasets by the ADASYN.

Fig. 13. The imbalanced and rebalanced datasets visualization in the PCA principle component space.

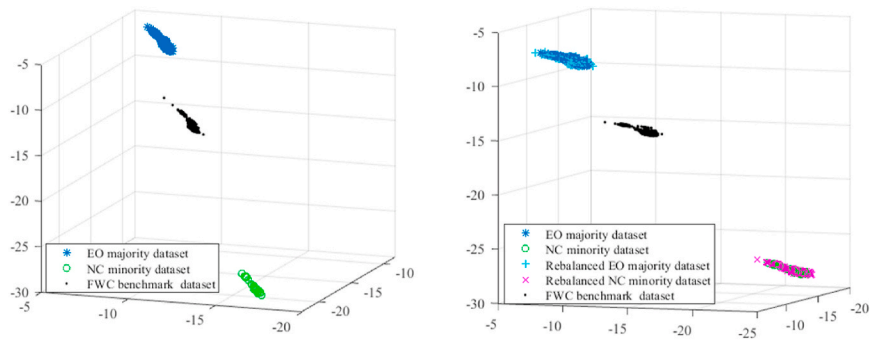
7.4.2. Comparisons for the severity degree Level-1

Compared with the other fault severity degrees, the faults corresponding to the severity degree Level-1 are more different to identify because they are incipient faults. Therefore, the imbalanced chiller datasets with different values of the index IR are applied to prove the superior fault diagnosis ability of the developed HRIELM based scheme.

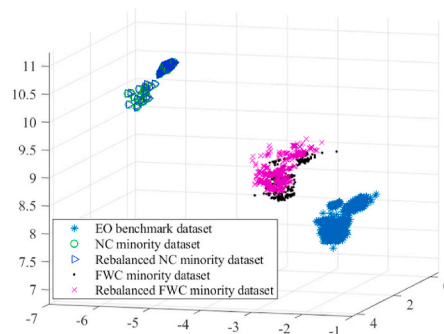
7.4.2.1. The fault diagnosis performance comparisons for $IR = 0.1$. The imbalanced dataset with $IR = 0.1$ of the severity degree Level-1 is first employed to evaluate the effectiveness of the HRIELM based scheme. The detailed sample numbers for this imbalanced dataset which consists of the seven diverse fault pattern subsets are given out in Table 3. According to the definition of the index IR , the minority and majority fault pattern subsets are randomly set up in imbalanced datasets with different index IR values. As exhibited in Table 3, the seven diverse fault pattern subsets own significantly different numbers of the fault samples. In particular, the fault pattern FEW subset contains 350 samples, which is the largest number of the fault samples. While the fault pattern EO subset has the smallest number of the fault samples, with only 36 samples. To settle the issue of data imbalance, we adopt the proposed HRT approach once to rebalance these seven imbalanced fault pattern subsets by treating the fault pattern CF subset as the benchmark dataset. Therefore, as shown in Table 3, every rebalanced fault pattern subset involves 205 fault samples. For a more intuitionistic comparison, the numbers of the samples in the imbalanced and rebalanced seven fault pattern subsets displayed in Table 3 are drawn as the bar charts in Fig. 15.

After the original imbalanced dataset is rebalanced utilizing the HRT by regarding each fault pattern subset as the benchmark dataset, the HRIELM based approach is further implemented to diagnose the fault patterns of the snapshot datasets. The fault diagnosis results of the HRIELM, BGELM, BTELM, WELM, and BWELM are calculated and exhibited in Fig. 16 and Table 4. Specifically, the confusion matrices of the five fault diagnosis approaches are displayed in Fig. 16.

As revealed in Fig. 16, the figures in the deep green squares denote the numbers of correctly identified fault snapshot samples, and the figures in the light green squares indicate the numbers of fault snapshot samples that are incorrectly identified. In the case of Fig. 16, the figures in the light green squares in the fourth column of Fig. 16(a), (b), (c), and (d) are much larger than those in the fourth column of Fig. 16(e). Therefore, BGELM, BTELM, WELM, and BWELM are more likely than the HRIELM to diagnose the fault pattern



(a) The original imbalanced datasets. (b) The imbalanced datasets and rebalanced datasets by the HRT.



(c) The imbalanced datasets and rebalanced datasets by the ADASYN.

Fig. 14. The imbalanced and rebalanced datasets visualization in the PCA residual space.

Table 3

Sample numbers in the imbalanced and rebalanced datasets for $IR = 0.1$.

Fault patterns	Sample number in the imbalanced dataset	Sample number in the rebalanced dataset
CF	205	205
EO	36	205
FWC	280	205
FEW	350	205
NC	280	205
RL	132	205
RO	67	205

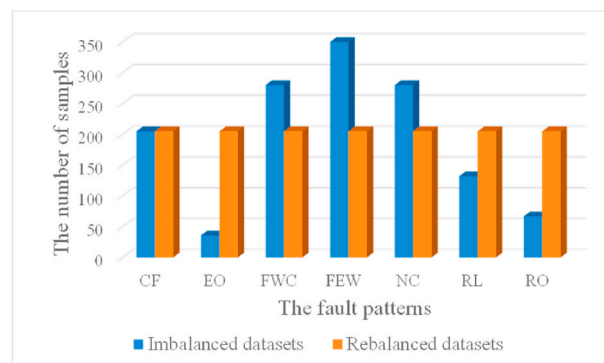


Fig. 15. The histogram of the sample numbers in the imbalanced and rebalanced datasets for $IR = 0.1$.

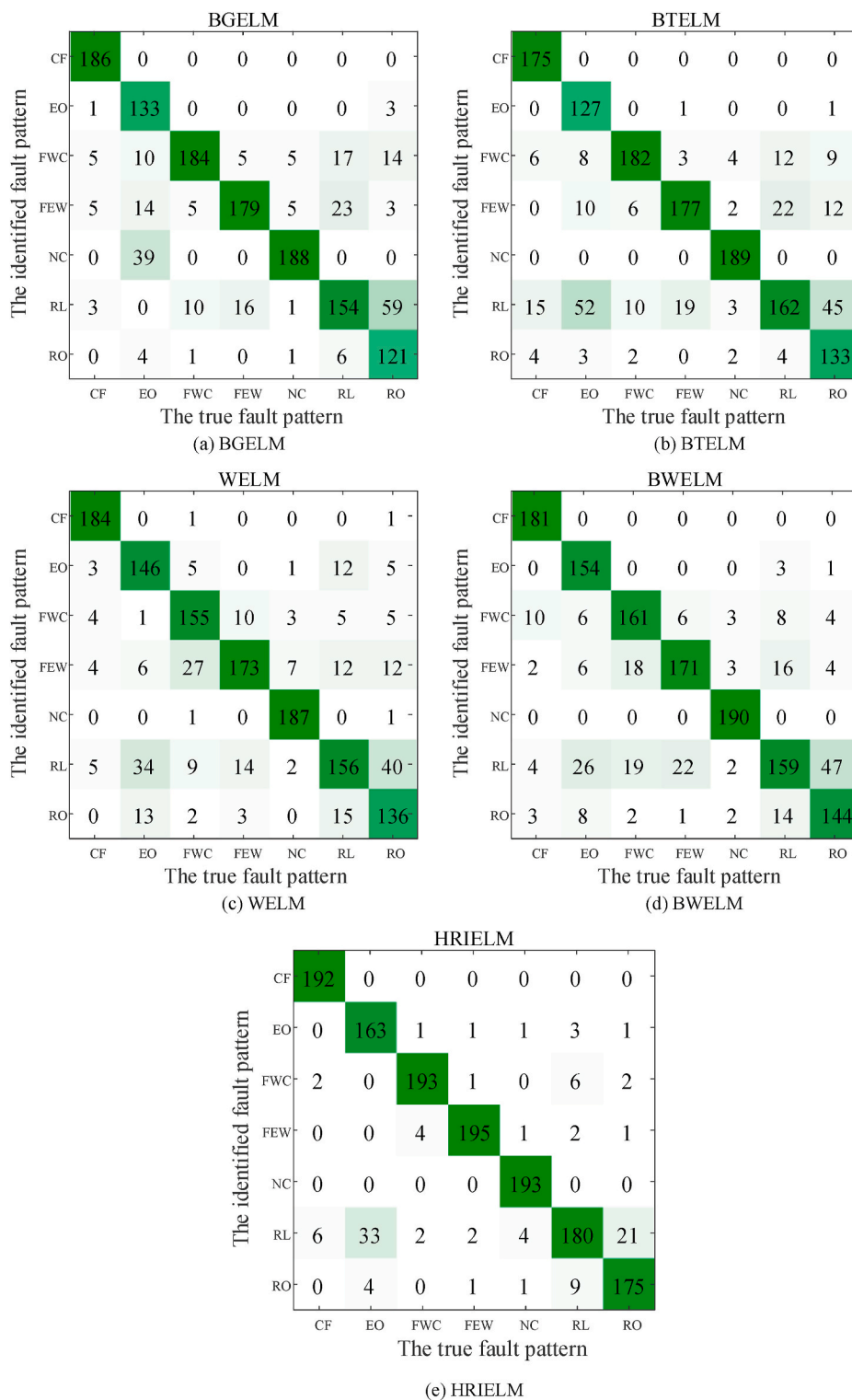


Fig. 16. The confusion matrices for the five fault diagnosis methods when $IR = 0.1$, (a) BGELM, (b) BTELM, (c) WELM, (d) BWELM, (e) HRIELM.

FEW as the other fault patterns by mistake. On the whole, Fig. 16(a), (b), (c), and (d) all exhibit more light green squares than Fig. 16(e), which demonstrates that the BGELM, BTELM, WELM, and BWELM incorrectly classified more snapshot samples of the seven fault patterns than the HRIELM.

Table 4Comparison of the index ACC values for the five methods under $IR = 0.1$.

Diagnosis methods	CF	EO	FWC	FEW	NC	RL	RO
BGELM	93.00%	66.50%	92.00%	89.50%	94.00%	77.00%	60.50%
BTELM	87.50%	63.50%	91.00%	88.50%	94.50%	81.00%	66.50%
WELM	92.00%	73.00%	77.50%	86.50%	93.50%	78.00%	68.00%
BWELM	90.50%	77.00%	80.50%	85.50%	95.00%	79.50%	72.00%
HRIELM	96.00%	81.50%	96.50%	97.50%	96.50%	90.00%	87.50%

The fault diagnosis accuracies of the HRIELM, BGELM, BTELM, WELM, and BWELM are further revealed in Table 4. As can be seen from Table 4, the index ACC values of the developed HRIELM are much larger than that of the other four approaches, especially for the fault patterns RO and EO which consist of the smallest numbers of the fault samples. Specifically, for the fault pattern RO, the value of the index ACC for the HRIELM method is 87.50%, which respectively increases by 27%, 21%, 19.5%, and 15.5% compared with that of the BGELM, BTELM, WELM, and BWELM. Similarly, the HRIELM's index ACC value (81.5%) for the fault pattern EO is respectively 15%, 18%, 8.5%, and 4.5% more than that of the BGELM, BTELM, WELM, and BWELM. For the convenience of comparison, the fault diagnosis accuracies of the five methods are visualized in Fig. 17. According to Fig. 17, the intuitionistic comparison proves the remarkable fault diagnosis effectiveness of the HRIELM over the BGELM, BTELM, WELM, and BWELM, in terms of identifying all the seven fault patterns.

7.4.2.2. The fault diagnosis performance comparisons for $IR = 0.6$. Another imbalanced dataset with $IR = 0.6$ is further used to measure the performance of the five approaches. Table 5 reveals the sample numbers in the seven imbalanced fault pattern subsets in detail. As shown in Table 5, the fault pattern CF subset possesses the largest number of fault samples, i.e., the 360 samples. Nevertheless, there are only 216 fault samples in the fault pattern FWC subset, which is the smallest number of the samples. Before conducting the fault diagnosis, the presented HRT method is employed to figure out the imbalanced chiller dataset based on the fault pattern RL subset. Hence, each of the seven rebalanced fault pattern subsets retains 280 fault samples. To make a vivid analysis, the sample numbers in the seven fault pattern subsets before and after performing the HRT are illustrated in Fig. 18.

The developed HRIELM based method is further conducted to recognize the fault patterns of the snapshot datasets after the HRT is performed seven times to acquire the seven diverse rebalanced datasets. The confusion matrices of the HRIELM, BGELM, BTELM, WELM, and BWELM are shown in Fig. 19. According to Fig. 19, the figure in the deep green square of the second column in Fig. 19(e) is much larger than those of the second column in Fig. 19(a), (b), (c), and (d). This scene means that the proposed HRIELM correctly discerns more fault snapshot samples of the fault pattern EO than the other four diagnosis methods. In an overall view, there are more light green squares in Fig. 19(a), (b), (c), and (d) than those in Fig. 19(e), which illustrates that compared with the HRIELM, more fault snapshots samples are wrongly divided into the other different fault patterns by the BGELM, BTELM, WELM, and BWELM.

The fault diagnosis accuracies of these five approaches are also given out in Table 6. Among the five methods, the HRIELM achieves the largest values of the index ACC to diagnose all seven fault patterns. For instance, the HRIELM's index ACC value for the minority class fault pattern FWC is 93.00%, which increases by 7%, 11.5%, 13.5%, and 10% in comparison with that of the BGELM, BTELM, WELM, and BWELM. To make an ulteriorly contrast, the fault diagnosis accuracies of the five identification approaches are represented graphically in Fig. 20. Therefore, the experimental results show that the presented HRIELM is superior over the BGELM, BTELM, WELM, and BWELM to identify the chiller's faults under the imbalanced data environment with $IR = 0.6$.

In our work, each fault pattern subset's sample number is randomly determined when the value of the index IR changes. For $IR = 0.1$, the sample number of each fault pattern is randomly selected between 36 and 350, while for $IR = 0.6$, each fault pattern's sample number is randomly chosen from 126 to 360. In the experiment, the sample numbers of the fault patterns CF, EO, NC, RL, and RO in the imbalanced dataset with $IR = 0.6$ are much larger than those in the imbalanced dataset with $IR = 0.1$. Hence, compared with the imbalanced dataset with $IR = 0.1$, the fault diagnosis accuracies of the above five fault patterns increase for the imbalanced dataset with $IR = 0.6$. However, when the value of the index IR rises from 0.1 to 0.6, the sample number of the fault pattern FWC reduces from

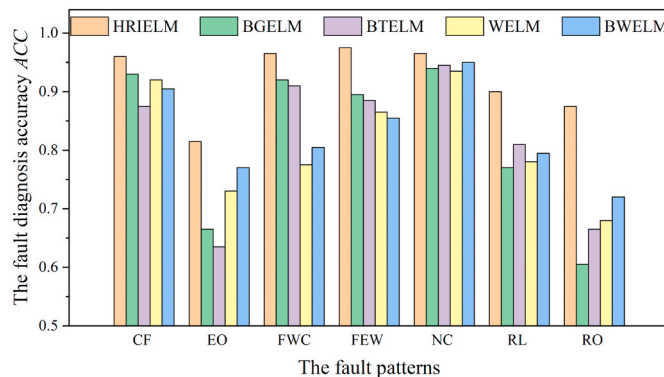
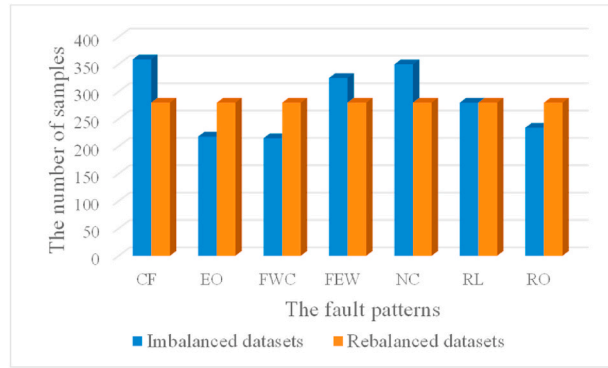
**Fig. 17.** The fault diagnosis accuracies of the five methods for $IR = 0.1$.

Table 5Sample numbers in the imbalanced and rebalanced datasets for $IR = 0.6$.

Fault patterns	Sample number in the imbalanced dataset	Sample number in the rebalanced dataset
CF	360	280
EO	218	280
FWC	216	280
FEW	325	280
NC	350	280
RL	280	280
RO	234	280

**Fig. 18.** The histogram of the sample numbers in the imbalanced and rebalanced datasets for $IR = 0.6$.

280 to 215 and the sample number of the fault pattern FEW decreases from 350 to 325. This leads to a slight reduction of the fault diagnosis accuracies for the fault patterns FWC and FEW in the imbalanced dataset with $IR = 0.6$.

7.4.2.3. Comparisons of the indices ACC_{total} and $G-mean$ for the $IR = 0.1$ and $IR = 0.6$. To further test the diagnosis effectiveness of the chiller under the imbalanced data environments, the values of overall fault diagnosis accuracy ACC_{total} and $G-mean$ for the HRIELM, BGELM, BTELM, WELM, and BWELM under the conditions $IR = 0.1$ and $IR = 0.6$ are listed in Table 7. Although the sample numbers in the seven fault patterns are much more imbalanced when $IR = 0.1$ than those when $IR = 0.6$, the differences between the values of the HRIELM's indices ACC_{total} and $G-mean$ are the smallest for $IR = 0.1$ and $IR = 0.6$ among the five approaches, which are respectively only 0.17% and 0.03%. Nevertheless, the differences between the indices ACC_{total} and $G-mean$ values are much larger for the BGELM, BTELM, WELM, and BWELM. Specifically, these differences are 1.09%, 0.84%, 0.49%, and 0.33% for $IR = 0.1$ and 0.12%, 0.15%, 0.51%, and 0.19% for $IR = 0.6$. The slight differences between the HRIELM's indices ACC_{total} and $G-mean$ for various IR values prove the superiority of the HRIELM for implementing the imbalanced data based chiller fault diagnosis.

To make a further comparison, the line charts of the indices ACC_{total} and $G-mean$ for the five fault diagnosis approaches under the conditions $IR = 0.1$ and $IR = 0.6$ are exhibited in Fig. 21. As can be seen from Fig. 21, the values of indices ACC_{total} and $G-mean$ of the five methods are significant improved when $IR = 0.6$ compared with those when $IR = 0.1$. To be specific, because the sample numbers in the seven fault patterns are seriously imbalanced when $IR = 0.1$, the values of the index ACC_{total} for the BGELM, BTELM, WELM, and BWELM are very different from the values of the index $G-mean$. While the index ACC_{total} value of the suggested HRIELM is still similar to the index $G-mean$ value, due to the HRIELM's excellent capability of managing the imbalanced data. On the contrary, the sample numbers in the seven fault patterns are relatively balanced when $IR = 0.6$. Consequently, the values of the index ACC_{total} for the five methods are close to the values of the index $G-mean$. In particular, the HRIELM's index ACC_{total} value is almost completely coincided with the value of its $G-mean$.

7.4.2.4. Comparisons of the fault diagnosis results with IR varying from 0.1 to 1. To inspect the fault identification effectiveness of the presented HRIELM method under different data imbalance rates, the recognition results of the HRIELM, BGELM, BTELM, WELM, and BWELM for the multiple imbalanced datasets with the index IR varying from 0.1 to 1 are computed and displayed in Table 8 and Fig. 22. As revealed in Table 8, when the IR changes from 0.1 to 1, the HRIELM's ACC_{total} values are all above 92.00% and they unfold the smallest variations among the five methods. In particular, the biggest distinction of the HRIELM's ACC_{total} with the varying values of IR is calculated as only 3.08%. Whereas the values of ACC_{total} for the BGELM, BTELM, WELM, and BWELM are respectively all below 87.70%, 87.20%, 88.10%, and 90.90% and they uncover much larger variations than those of the HRIELM when the IR changes from 0.1 to 1. Particularly, the biggest distinctions of the index ACC_{total} values for the BGELM, BTELM, WELM, and BWELM are respectively 17.07%, 13.28%, 14.86%, and 14.72% with the IR varying. Meanwhile, during the IR changing from 0.1 to 1, the HRIELM's $G-mean$ values are always the largest and they also display the smallest variations among the five fault diagnosis methods. To be specific, when the value of IR varies, the values of the $G-mean$ for the HRIELM are all above 91.9% and the biggest distinction of the $G-mean$ is computed to be only 3.34%.

Furthermore, the values of the ACC_{total} and $G-mean$ for the five approaches under different IR values are visualized in Fig. 22, where

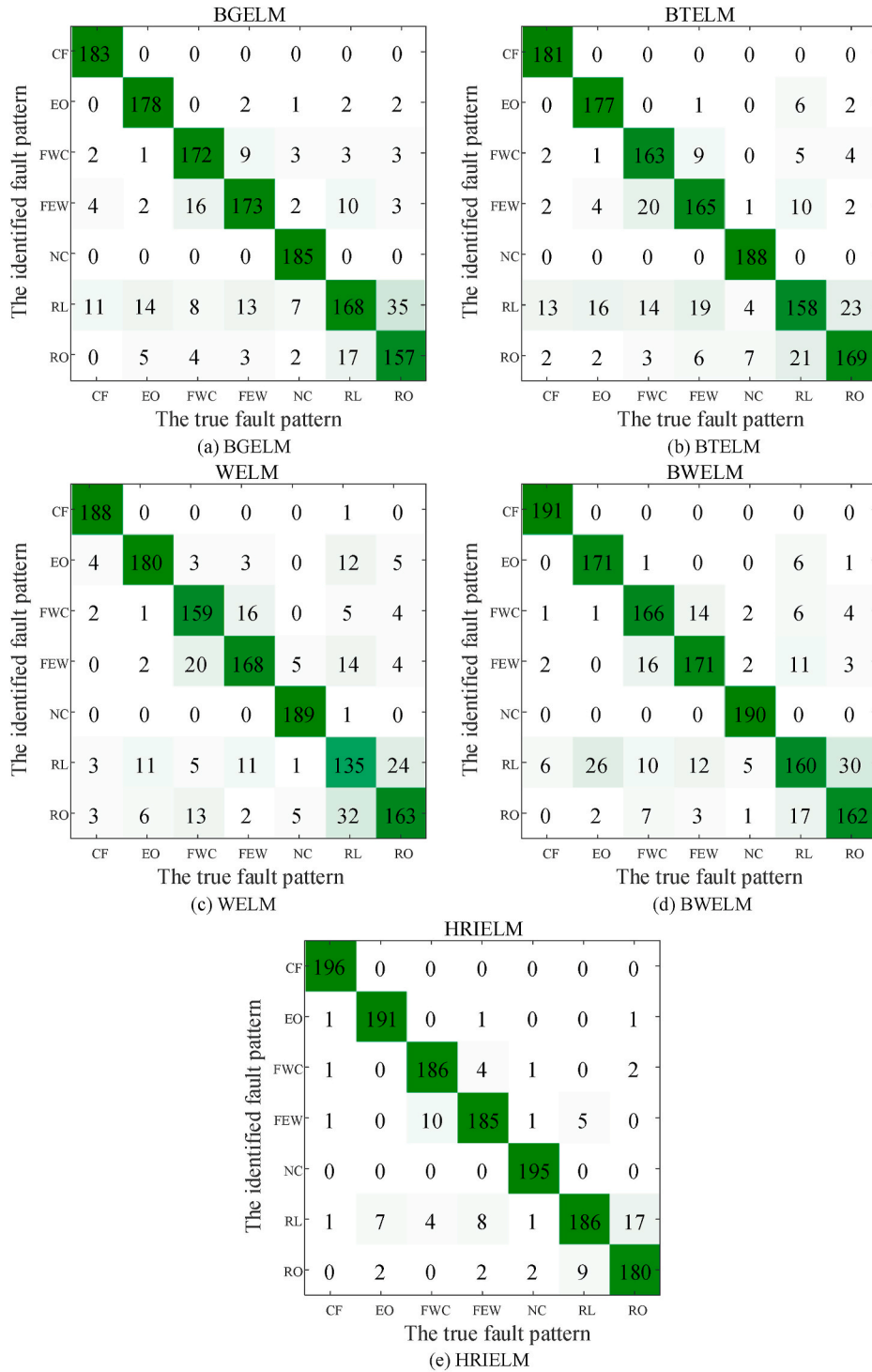


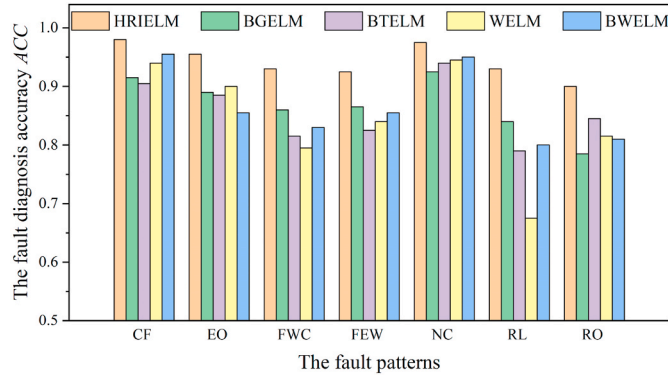
Fig. 19. The confusion matrices for the five fault diagnosis methods when $IR = 0.6$, (a) BGELM, (b) BTELM, (c) WELM, (d) BWELM, (e) HRIELM.

the HRIELM's ACC_{total} values are denoted as the block square line and the HRIELM's $G-mean$ values are represented as the red dot line. It can be seen from Fig. 22(a) and (b), the lines of the HRIELM's ACC_{total} and $G-mean$ are both above the corresponding lines of the other four approaches. Thus, the HRIELM is more efficient to diagnose the chiller fault under different data imbalance rates than the BGELM, BTELM, WELM, and BWELM.

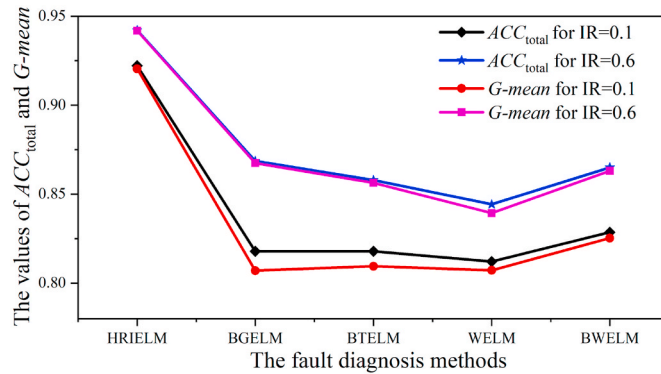
In addition, according to Fig. 22, the indices ACC_{total} and $G-mean$ values of the five fault diagnosis methods suddenly decrease for

Table 6The comparison of the index ACC values for the five methods under $IR = 0.6$.

Diagnosis methods	CF	EO	FWC	FEW	NC	RL	RO
BGELM	91.50%	89.00%	86.00%	86.50%	92.50%	84.00%	78.50%
BTELM	90.50%	88.50%	81.50%	82.50%	94.00%	79.00%	84.50%
WELM	94.00%	90.00%	79.50%	84.00%	94.50%	67.50%	81.50%
BWELM	95.50%	85.50%	83.00%	85.50%	95.00%	80.00%	81.00%
HRIELM	98.00%	95.50%	93.00%	92.50%	97.50%	93.00%	90.00%

**Fig. 20.** The identification accuracies of the five approaches for $IR = 0.6$.**Table 7**The comparison of the indices ACC_{total} and G-mean for the five methods when $IR = 0.1$ and $IR = 0.6$.

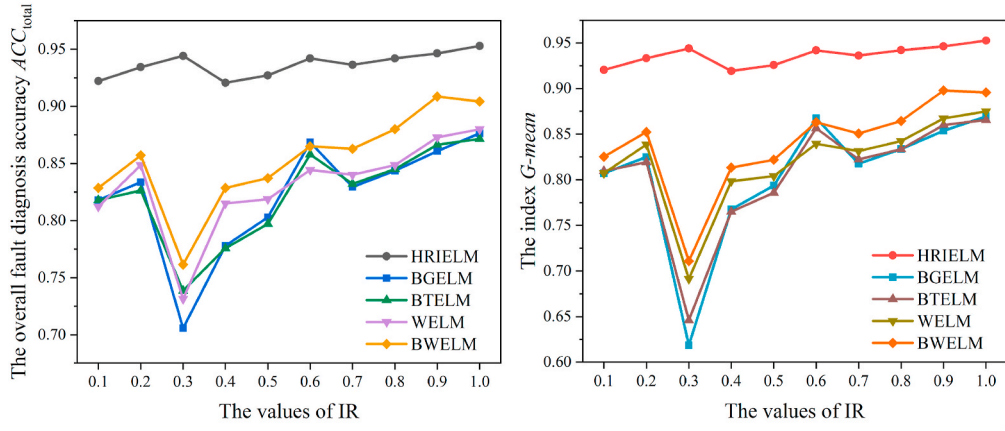
Diagnosis methods	$IR = 0.1$		$IR = 0.6$	
	ACC_{total}	G-mean	ACC_{total}	G-mean
BGELM	81.79%	80.70%	86.86%	86.74%
BTELM	81.79%	80.95%	85.79%	85.64%
WELM	81.21%	80.72%	84.43%	83.94%
BWELM	82.86%	82.53%	86.50%	86.31%
HRIELM	92.21%	92.04%	94.21%	94.18%

**Fig. 21.** The line charts of the indices ACC_{total} and G-mean for the five methods when $IR = 0.1$ and $IR = 0.6$.

$IR = 0.3$ and then gradually increase again when IR varies from 0.4 to 1. This phenomenon results from the numbers of samples in various fault pattern subsets that are randomly determined from different numerical intervals for $IR \leq 0.3$ and $IR \geq 0.4$ in our work. Table 9 gives out the different numerical intervals applied to randomly select the sample numbers of the diverse fault pattern subsets in an imbalanced dataset when IR changes from 0.1 to 1. As exhibited in Table 9, with the value of IR increases from 0.1 to 0.3, the numerical interval becomes smaller and smaller. In particular, when $IR = 0.3$, the numerical interval decreases to the minimum [13, 15]. This means that when $IR = 0.3$, the fault pattern subsets in an imbalanced dataset would have the minimum sample numbers compared with those under the other different IR values. Therefore, for $IR = 0.3$, the limited sample numbers of the fault pattern

Table 8The comparison of the indices ACC_{total} and $G-mean$ for the five methods with IR varying from 0.1 to 1.

IR	BGELM		BTLM		WELM		BWELM		HRIELM	
	ACC_{total}	$G-mean$	ACC_{total}	$G-mean$	ACC_{total}	$G-mean$	ACC_{total}	$G-mean$	ACC_{total}	$G-mean$
0.1	81.79%	80.70%	81.79%	80.95%	81.21%	80.72%	82.86%	82.53%	92.21%	92.04%
0.2	83.36%	82.51%	82.64%	81.92%	84.86%	83.85%	85.71%	85.23%	93.43%	93.32%
0.3	70.57%	61.86%	73.86%	64.60%	73.14%	69.14%	76.14%	71.09%	94.43%	94.39%
0.4	77.79%	76.74%	77.57%	76.50%	81.50%	79.82%	82.86%	81.33%	92.07%	91.92%
0.5	80.29%	79.35%	79.71%	78.58%	81.86%	80.39%	83.71%	82.19%	92.71%	92.57%
0.6	86.86%	86.74%	85.79%	85.64%	84.43%	83.94%	86.50%	86.31%	94.21%	94.18%
0.7	82.93%	81.75%	83.21%	82.23%	84.00%	83.14%	86.29%	85.07%	93.64%	93.62%
0.8	84.36%	83.38%	84.50%	83.35%	84.86%	84.25%	88.00%	86.44%	94.21%	94.20%
0.9	86.07%	85.36%	86.64%	86.01%	87.29%	86.72%	90.86%	89.79%	94.64%	94.62%
1	87.64%	86.94%	87.14%	86.56%	88.00%	87.49%	90.43%	89.57%	95.29%	95.26%

(a) The variation tendency of the ACC_{total} . (b) The variation tendency of the $G-mean$.**Fig. 22.** The line charts of the indices ACC_{total} and $G-mean$ for the five methods with IR varying from 0.1 to 1.**Table 9**The intervals utilized to randomly determine the sample numbers under different IR values.

IR	The interval [min (N_i), max (N_i)]
0.1	[36, 350]
0.2	[36, 175]
0.3	[36, 88]
0.4	[36, 117]
0.5	[180, 360]
0.6	[216, 360]
0.7	[252, 360]
0.8	[288, 360]
0.9	[324, 360]
1	[359, 360]

subsets result in the ACC_{total} and $G-mean$ values of the five methods abruptly decreasing. On the contrary, with IR raising from 0.4 to 1, the sample numbers of the fault pattern subsets in an imbalanced dataset go on growing. This situation brings about the ACC_{total} and $G-mean$ values increasing by degrees when IR changes from 0.4 to 1.

7.4.3. Comparisons for the severity degrees Level-2 ~ Level-4

The recognition results of the HRIELM, BGELM, BTLM, WELM, and BWELM based approaches for the other three fault severity levels, i.e., Level-2, Level-3, and Level-4, are further estimated to examine the performance influences of different fault severity levels. As shown in Fig. 23, for the three fault severity levels, the indices ACC_{total} and $G-mean$ values of the five fault diagnosis approaches all fluctuate up and down along with different IR values. Although the values of the HRIELM's diagnosis performance indices slightly decline for $IR = 0.3$, the HRIELM obtains the largest and the most stable ACC_{total} and $G-mean$ values among the five methods for the fault severity degrees Level-2, Level-3, and Level-4. Again, the visualized comparisons of fault diagnosis results exhibited in Fig. 23 prove the excellent fault diagnosis capability of the HRIELM, in terms of different fault severity levels and varying data imbalance

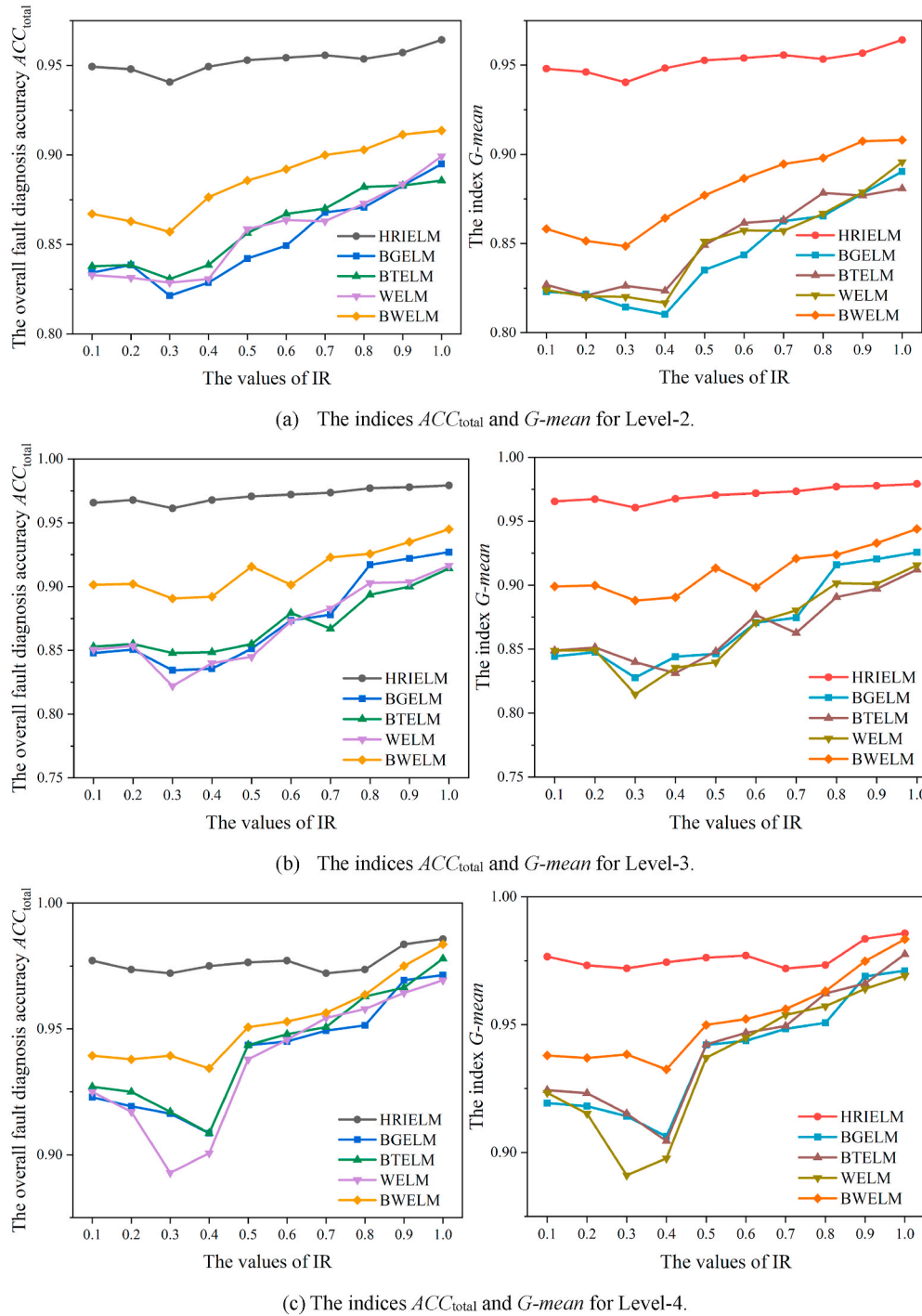


Fig. 23. The ACC_{total} and $G-mean$ values of the five methods for different fault severity levels.

rates.

8. Conclusions

A novel HRIELM based scheme is suggested to diagnose the chiller fault under the imbalanced data environments in our work. As far as we know, we are the first to combine a new hybrid resampling technique with an improved selective ensemble ELM model to identify the faults when the sample numbers of the historical fault pattern datasets are imbalanced. To be specific, our work has three

contributions. Firstly, the HRT technique is proposed to resample and rebalance the chiller's imbalanced historical fault pattern datasets, by fusing the SMOTE over-sampling method with the random under-sampling method. Secondly, to alleviate the computational complexity and lay a foundation for the subsequent modified selective ensemble learning strategy, the ELM is utilized as the basic fault diagnosis model and a number of basic ELM diagnosis models are established on the multiple different rebalanced training datasets. At last, an improved selective ensemble learning strategy is developed to diagnose the fault pattern of snapshot samples by picking out and weighting the basic ELM models with superior performance, to further enhance the fault identification effectiveness of the chiller and reduce the computational burden.

Detailed experiments and comparisons implemented on the ASHRAE RP-1043 experimental datasets verify the excellent performance of the proposed HRIELM scheme for the imbalanced data based chiller fault diagnosis. To be specific, in comparison with the related adaptive synthetic (ADASYN) sampling approach, the developed HRT resampling technique reveals superior effectiveness to deal with the problem of data imbalance by enlarging the minority classes and shrinking the majority classes. In addition, compared with the relevant BGELM, BTELM, WELM, and BWELM based methods on four different severity levels imbalanced datasets with the imbalance rate (*IR*) varying from 0.1 to 1, the experiments demonstrate the outstanding diagnosis performance and remarkable improvements of the proposed HRIELM based fault diagnosis scheme in terms of the three evaluation indices: fault diagnosis accuracy, overall fault diagnosis accuracy, and geometric mean.

In practice, it is always time-consuming and laborious to gather sufficient labeled operation data for the chiller's different fault patterns, while large amounts of the chiller's unlabeled operation data are more easily to be collected. To tackle the tough issue of imbalanced labeled and unlabeled operation data during recognizing fault, further studies are suggested to investigate some semi-supervised learning based methods, such as the semi-supervised ELM, semi-supervised deep neural networks, and so on, to build more reliable chiller fault diagnosis models.

Author statement

Hanyuan Zhang*: Investigation, Methodology, Project administration, Supervision, Funding acquisition. Wenxin Yang: Conceptualization, Investigation, Software. Weilin Yi: Methodology, Investigation, Software. Jit Bing Lim: Conceptualization, Methodology, Writing – original draft. Zenghui An: Data curation, Conceptualization, Writing – original draft. Chengdong Li: Software, Validation, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

Data will be made available on request.

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References

- [1] J. Chen, L. Zhang, Y. Li, et al., A review of computing-based automated fault detection and diagnosis of heating, ventilation and air conditioning systems, *Renew. Sustain. Energy Rev.* 161 (2022), 112395, <https://doi.org/10.1016/j.rser.2022.112395>.
- [2] M.S. Mirnaghi, F. Haghighat, Fault detection and diagnosis of large-scale HVAC systems in buildings using data-driven methods: a comprehensive review, *Energy Build.* 229 (2020), 110492, <https://doi.org/10.1016/j.enbuild.2020.110492>.
- [3] Z. Li, T. Zheng, Y. Wang, et al., A novel method for imbalanced fault diagnosis of rotating machinery based on generative adversarial networks, *IEEE Trans. Instrum. Meas.* 70 (2020) 1–17, <https://doi.org/10.1109/TIM.2020.3009343>.
- [4] Ke Yan, Chong Adrian, Yuchang Mo, Generative adversarial network for fault detection diagnosis of chillers, *Build. Environ.* 172 (2020), 106698, <https://doi.org/10.1016/j.buildenv.2020.106698>.
- [5] Cheng Fan, Xueqing Li, Yang Zhao, Jiayuan Wang, Quantitative assessments on advanced data synthesis strategies for enhancing imbalanced AHU fault diagnosis performance, *Energy Build.* 252 (2021), 111423, <https://doi.org/10.1016/j.enbuild.2021.111423>.
- [6] J. Chen, H. Huang, A.G. Cohn, et al., Machine learning-based classification of rock discontinuity trace: SMOTE oversampling integrated with GBT ensemble learning, *Int. J. Min. Sci. Technol.* 32 (2) (2022) 309–322, <https://doi.org/10.1016/j.ijmst.2021.08.004>.
- [7] A.M. Zhang, H.L. Yu, S.L. Zhou, Z.J. Huan, X.B. Yang, Instance weighted SMOTE by indirectly exploring the data distribution, *Knowl. Base Syst.* 249 (2022), 108919, <https://doi.org/10.1016/j.knsys.2022.108919>.
- [8] S. Lu, P. Zhou, X. Wang, et al., Condition monitoring and fault diagnosis of motor bearings using undersampled vibration signals from a wireless sensor network, *J. Sound Vib.* 414 (2018) 81–96, <https://doi.org/10.1016/j.jsv.2017.11.007>.
- [9] M. Kabir, S. Ahmad, M. Iqbal, et al., Improving prediction of extracellular matrix proteins using evolutionary information via a grey system model and asymmetric under-sampling technique, *Chemometr. Intell. Lab. Syst.* 174 (2018) 22–32, <https://doi.org/10.1016/j.chemolab.2018.01.004>.
- [10] Q. Hu, A. Qin, Q. Zhang, et al., Fault diagnosis based on weighted extreme learning machine with wavelet packet decomposition and KPCA, *IEEE Sensor. J.* 18 (20) (2018) 8472–8483, <https://doi.org/10.1109/JSEN.2018.2866708>.
- [11] Y.Q. Chen, O. Fink, G. Sansavini, Combined fault location and classification for power transmission lines fault diagnosis with integrated feature extraction, *IEEE Trans. Ind. Electron.* 65 (1) (2017) 561–569, <https://doi.org/10.1109/TIE.2017.2721922>.

- [12] J. Liu, J. He, W. Zhang, et al., ANID-SEoKELM: adaptive network intrusion detection based on selective ensemble of kernel ELMs with random features, *Knowl. Base Syst.* 177 (2019) 104–116, <https://doi.org/10.1016/j.knsys.2019.04.008>.
- [13] C. Perales-Gonzalez, M. Carbonero-Ruz, D. Becerra-Alonso, et al., Regularized ensemble neural networks models in the Extreme Learning Machine framework, *Neurocomputing* 361 (2019) 196–211, <https://doi.org/10.1016/j.neucom.2019.06.040>.
- [14] S.X. Lv, L. Peng, H. Hu, et al., Effective machine learning model combination based on selective ensemble strategy for time series forecasting, *Inf. Sci.* 612 (2022) 994–1023, <https://doi.org/10.1016/j.ins.2022.09.002>.
- [15] H. Jin, Y. Li, B. Wang, et al., Adaptive forecasting of wind power based on selective ensemble of offline global and online local learning, *Energy Convers. Manag.* 271 (2022), 116296, <https://doi.org/10.1016/j.enconman.2022.116296>.
- [16] X. Li, H. Jiang, M. Niu, et al., An enhanced selective ensemble deep learning method for rolling bearing fault diagnosis with beetle antennae search algorithm, *Mech. Syst. Signal Process.* 142 (2020), 106752, <https://doi.org/10.1016/j.ymssp.2020.106752>.
- [17] T. Zhang, J. Chen, F. Li, et al., Intelligent fault diagnosis of machines with small & imbalanced data: a state-of-the-art review and possible extensions, *ISA Trans.* 119 (2022) 152–171, <https://doi.org/10.1016/j.isatra.2021.02.042>.
- [18] A.P. Rogers, F. Guo, B.P. Rasmussen, A review of fault detection and diagnosis methods for residential air conditioning systems, *Build. Environ.* 161 (2019), 106236, <https://doi.org/10.1016/j.buildenv.2019.106236>.
- [19] J. Park, J.M. Ha, H. Oh, et al., Model-based fault diagnosis of a planetary gear: a novel approach using transmission error, *IEEE Trans. Reliab.* 65 (4) (2016) 1830–1841, <https://doi.org/10.1109/TR.2016.2590997>.
- [20] Zhen Zhao, , Peter Xiaoping Liu, Jinfeng Gao, Model-based fault diagnosis methods for systems with stochastic process -A survey, *Neurocomputing* 513 (2022) 137–152, <https://doi.org/10.1016/j.neucom.2022.09.134>.
- [21] Vijay Singh, Jyotirmay Mathur, Aviruch Bhatia, A comprehensive review: fault detection, diagnostics, prognostics, and fault modeling in HVAC systems, *Int. J. Refrig.* 144 (2022) 283–295, <https://doi.org/10.1016/j.ijrefrig.2022.08.017>.
- [22] J. Wen, H. Yao, Z. Ji, et al., On fault diagnosis for high-g accelerometers via data-driven models, *IEEE Sensor. J.* 21 (2) (2020) 1359–1368, <https://doi.org/10.1109/JSEN.2020.3019632>.
- [23] S. Asgari, R. Gupta, I.K. Puri, et al., A data-driven approach to simultaneous fault detection and diagnosis in data centers, *Appl. Soft Comput.* 110 (2021), 107638, <https://doi.org/10.1016/j.asoc.2021.107638>.
- [24] A. Taal, L. Itard, P&ID-based automated fault identification for energy performance diagnosis in HVAC systems: 4S3F method, development of DBN models and application to an ATEs system, *Energy Build.* 224 (2020), 110289, <https://doi.org/10.1016/j.enbuild.2020.110289>.
- [25] Y. Fan, X. Cui, H. Han, et al., Feasibility and improvement of fault detection and diagnosis based on factory-installed sensors for chillers, *Appl. Therm. Eng.* 164 (2020), 114506, <https://doi.org/10.1016/j.applthermaleng.2019.114506>.
- [26] M. Bonvini, M.D. Sohn, J. Granderson, et al., Robust on-line fault detection diagnosis for HVAC components based on nonlinear state estimation techniques, *Appl. Energy* 124 (2014) 156–166, <https://doi.org/10.1016/j.apenergy.2014.03.009>.
- [27] R. Van de Sand, S. Corasaniti, J. Reiff-Stephan, Data-driven fault diagnosis for heterogeneous chillers using domain adaptation techniques, *Control Eng. Pract.* 112 (2021), 104815, <https://doi.org/10.1016/j.conengprac.2021.104815>.
- [28] D. Li, G. Hu, C.J. Spanos, A data-driven strategy for detection and diagnosis of building chiller faults using linear discriminant analysis, *Energy Build.* 128 (2016) 519–529, <https://doi.org/10.1016/j.enbuild.2016.07.014>.
- [29] J. Liu, D. Shi, G. Li, et al., Data-driven and association rule mining-based fault diagnosis and action mechanism analysis for building chillers, *Energy Build.* 216 (2020), 109957, <https://doi.org/10.1016/j.enbuild.2020.109957>.
- [30] Hanyuan Zhang, C. Li, D. Li, et al., Fault detection and diagnosis of the air handling unit via an enhanced kernel slow feature analysis approach considering the time-wise and batch-wise dynamics, *Energy Build.* 253 (2021), 111467, <https://doi.org/10.1016/j.enbuild.2021.111467>.
- [31] Hanyuan Zhang, C. Li, Q. Wei, et al., Fault detection and diagnosis of the air handling unit via combining the feature sparse representation based dynamic SFA and the LSTM network, *Energy Build.* (2022), 112241, <https://doi.org/10.1016/j.enbuild.2022.112241>.
- [32] C. Li, C. Shen, Hanyuan Zhang, et al., A novel temporal convolutional network via enhancing feature extraction for the chiller fault diagnosis, *J. Build. Eng.* 42 (2021), 103014, <https://doi.org/10.1016/j.jobbe.2021.103014>.
- [33] Y. Yu, Hanyuan Zhang, W. Peng, et al., Images based fault diagnosis of air handling unit via combining kernel slow feature analysis and deep learning method, *J. Build. Eng.* (2022), 104718, <https://doi.org/10.1016/j.jobbe.2022.104718>.
- [34] Z. An, X. Jiang, R. Yang, Hanyuan Zhang, et al., Actively imaginative data augmentation for machinery diagnosis under large speed fluctuation conditions, *IEEE Trans. Ind. Inf.* (2023), <https://doi.org/10.1109/TII.2022.3218737> in press.
- [35] K. Xu, S. Li, X. Jiang, et al., A novel transfer diagnosis method under unbalanced sample based on discrete-peak joint attention enhancement mechanism, *Knowl. Base Syst.* 212 (2021), 106645, <https://doi.org/10.1016/j.knsys.2020.106645>.
- [36] N. Wang, Y. Zhang, X. Zhao, et al., Search-based Cost-Sensitive Hypergraph Learning for Anomaly Detection, *Information Sciences*, 2022, <https://doi.org/10.1016/j.ins.2022.07.029>.
- [37] F. Jiang, X. Yu, J. Du, et al., Ensemble learning based on approximate reducts and bootstrap sampling, *Inf. Sci.* 547 (2021) 797–813, <https://doi.org/10.1016/j.ins.2020.08.069>.
- [38] J. Wang, Y. Zhang, F. Zhang, et al., Accuracy-improved bearing fault diagnosis method based on AVMD theory and AWPSO-ELM model, *Measurement* 181 (2021), 109666, <https://doi.org/10.1016/j.measurement.2021.109666>.
- [39] K. Yan, Z. Ji, H. Lu, et al., Fast and accurate classification of time series data using extended ELM: application in fault diagnosis of air handling units, *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 49 (7) (2017) 1349–1356, <https://doi.org/10.1109/TSMC.2017.2691774>.
- [40] Cunxiao Shen, Hanyuan Zhang, Songping Meng, Chengdong Li, Augmented data driven self-attention deep learning method for imbalanced fault diagnosis of the HVAC chiller, *Eng. Appl. Artif. Intell.* 117 (2023), 105540, <https://doi.org/10.1016/j.engappai.2022.105540>.
- [41] Z. Zhou, H. Chen, G. Li, et al., Data-driven fault diagnosis for residential variable refrigerant flow system on imbalanced data environments, *Int. J. Refrig.* 125 (2021) 34–43, <https://doi.org/10.1016/j.ijrefrig.2021.01.009>.
- [42] G.B. Huang, Q.Y. Zhu, C.K. Siew, Extreme learning machine: theory and applications, *Neurocomputing* 70 (1–3) (2006) 489–501, <https://doi.org/10.1016/j.neucom.2005.12.126>.
- [43] P. Li, Z. Liu, B. Anduv, et al., Diagnosis for multiple faults of chiller using ELM-KNN model enhanced by multi-label learning and specific feature combinations, *Build. Environ.* 214 (2022), 108904, <https://doi.org/10.1016/j.buildenv.2022.108904>.
- [44] M.C. Comstock, J.E. Braun, Development of Analysis Tools for the Evaluation of Fault Detection and Diagnostics in Chillers, ASHRAE Research Project RP-1043, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, 1999, pp. 20–99.
- [45] J. Liu, D. Shi, G. Li, Y. Xie, K. Li, et al., Data-driven and association rule mining-based fault diagnosis and action mechanism analysis for building chillers, *Energy Build.* 216 (2020), 109957, <https://doi.org/10.1016/j.enbuild.2020.109957>.
- [46] C.M. Vong, J. Du, C.M. Wong, et al., Postboosting using extended G-mean for online sequential multiclass imbalance learning, *IEEE Transact. Neural Networks Learn. Syst.* 29 (12) (2018) 6163–6177, <https://doi.org/10.1109/TNNLS.2018.2826553>.
- [47] Haibo He, Bai Yang, E.A. Garcia, Shutao Li Adasyn, Adaptive Synthetic Sampling Approach for Imbalanced Learning. 2008 IEEE International Joint Conference on Neural Networks, IEEE World Congress on Computational Intelligence), Hong Kong, 2008, pp. 1322–1328, <https://doi.org/10.1109/IJCNN.2008.4633969>.
- [48] Debaleena Datta, Pradeep Kumar Mallick, Annapareddy V.N. Reddy, et al., A hybrid classification of imbalanced hyperspectral images using ADASYN and enhanced deep subsampled multi-grained cascaded forest, *Rem. Sens.* 14 (19) (2022) 4853, <https://doi.org/10.3390/rs14194853>.